Calibration Behavior: 2024 vs. 2026

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Introduction

This report compares longitudinal calibration behavior between the 2024 and 2026 calibration datasets retrieved from RIPE Atlas Anchors. The analysis reproduces the drift analysis originally applied to the 2024 calibration data by Prof. Zach Weinberg of Carnegie Mellon University. For both the 2024 and 2026 data, calendar dates are preserved throughout the analysis. Thus, the `days` variable represents actual elapsed calendar days rather than the number of calibration files retained after missing days were removed.

Two measures of drift are examined:

1. **Maximum drift:** the largest instantaneous drift observed up to each point in time, with respect to the calibration on Day 0.
2. **Instantaneous drift:** the absolute proportional difference between an anchor's current calibration and its calibration on Day 0.

Load Data

The 2026 data requires us to calculate calibration value via given slope and intercept values.

The calibration value is calculated as:

$$y = 100m + b$$

where (m) is the slope and (b) is the intercept.

Table 1: Dataset Summaries

year	anchors	observations	first_date	last_date	calendar_days
2024	659	19111	2024-10-21	2024-11-18	28
2026	444	35520	2026-01-23	2026-04-14	81

Only anchors with complete observations across all retained calibration dates are used in the final comparison, despite the attempt to preserve all anchors with 0.8% retention across all files in the previous report *Anchor Representation: 2024 vs. 2026* available in the repository.

Calculating Drift

The following calculation is equivalent to Professor Weinberg's original 2024 analysis.

For each anchor:

$$drift_t = \frac{|calibration_t - calibration_0|}{calibration_0}$$

Overview

The report is organized as follows:

1. Maximum Drift
    1. 90-99.9th Percentile Anchor Summary
    2. Geographic Context of IDs Associated with repeated ASNs in the 90-99.9 Range
    3. Visualizing Repeated ASN in 90-99.9th Percentile Range
    4. Flaw with Maximum Calibration Drift Visualization: Repeated ASN Behavior and Day 0 as Ground Truth
    5. Geographic Context of Single Anchor Associated ASNs in the 90-99.9 Range
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    1. 90-99.9th Percentile Anchor Summary
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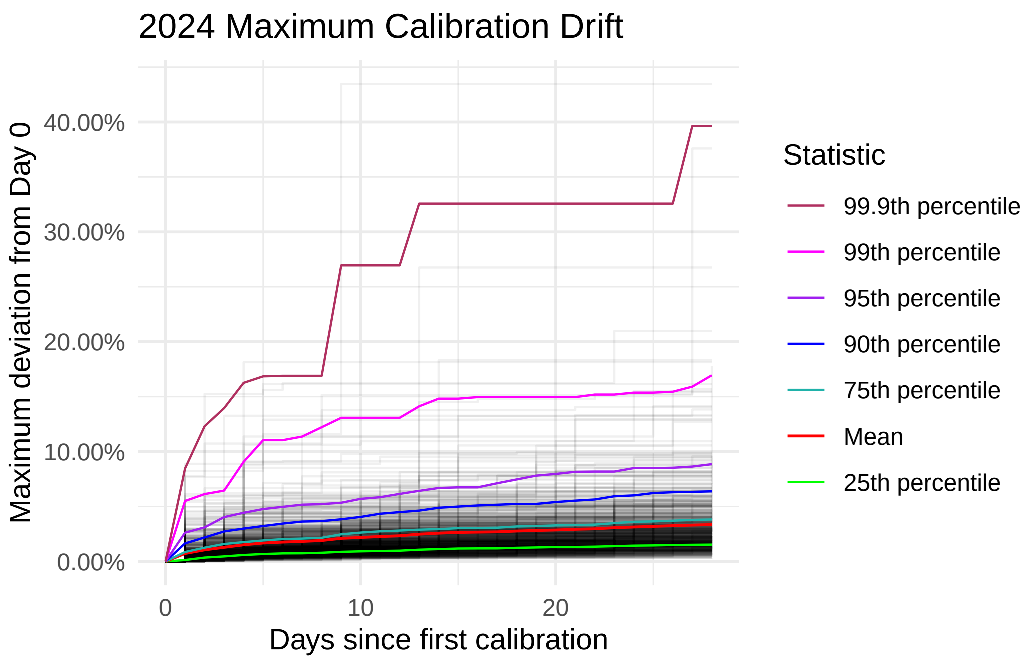
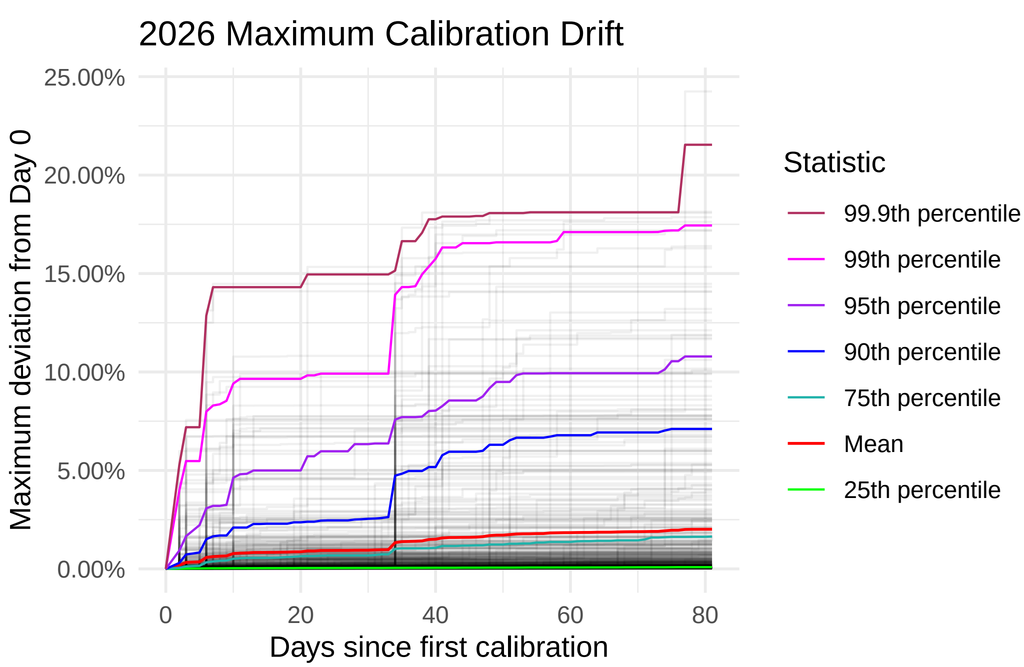
    3. Visualizing Repeated ASN in 90-99.9th Percentile Range
    4. Flaw with Maximum Calibration Drift Visualization: Repeated ASN Behavior and Day 0 as Ground Truth
    5. Geographic Context of Single Anchor Associated ASNs in the 90-99.9 Range
    6. Filtered Instantaneous Drift (Day 0 Adjustment: IDs from Instantaneous Drift 90-99.9th percentile)
    7. Discussion on Adjusted Instantaneous Drift Visualization
4. Instantaneous Drift Threshold
    1. Observations/Discussion
    2. Anchor IDs that Exceeded 20%, 10%, and 5%
5. Direct 2024 vs. 2026 Comparison
    1. Maximum Calibration Drift
    2. Instantaneous Calibration Drift
    3. Discussion
6. Notes for Interpretation/Process

3. Maximum Drift





3.1 90-99.9th Percentile Anchor Summary

- 653 anchors were in the range at least once in 2024 data
- 444 anchors were in the range at least once in 2026 data
- 276 anchors were in the range at least once in both 2024 and 2026 data
- 185 anchors occupied the range only one time each in both 2024 and 2026 data
- 91 anchors occupied the range more than one time each between 2024 and 2026 data

3.2 Geographic Context of IDs Associated with repeated ASNs in the 90-99.9 Range

Below, we take a look at the 91 IDs that appeared in the 90-99.9th percentile range more than one time each between the 2024 and 2026 datasets. After joining geographic context to these anchors `cleaned_persistent_90_999_both`, we find that several ASN IDs, cities, and countries are associated more than once with Anchor IDs consistently in the 90-99.9th percentile range. Below are tables for which there was more than anchor association.

Table 2: ASN Frequency

asn	anchors
14061	4
16276	4
3491	3
57695	3
36236	2
197540	2

AS14061 belongs to DigitalOcean, LLC, registered in the United States, and hosts 49.9% of its IPv4 share in the United States, 12.4% in Germany, 10.8% in the Netherlands, and 10.3% in Singapore. The rest of the percentages reported are below 10% and are reported to be in the United Kingdom, India, Canada, and Australia. The same countries are listed with different percentages for IPv6. Link for ASN 14061

AS16276 belongs to OVH SAS, registered in France. It hosts 55.5% of its IPv4 share in France, 17.5% in Canada, and less than 10% in the United States, Germany, the United Kingdom, Poland, Singapore, and Australia. It hosts less than 1% in India, Italy Ireland, Netherlands, Belgium, and Hong Kong. The same countries are listed for IPv6, with different percentages. Link for ASN 16276

AS3491 belongs to PCCW Global (HK) Ltd., registered in Hong Kong. It hosts 32% of its IPv6 share in Hong Kong, 28.9% in South Africa, 28.6% in Mozambique, and less than 3% in the United States, Singapore, Japan, and <1% in Germany, France, Taiwan, South Korea, United Kingdom, Australia, Netherlands, Mexico, Sweden, and Kenya. The same countries occupy the shares of IPv4 addresses, with different percentages. Link for ASN 3491

AS57695 belongs to Misaka Network, Inc., and is registered in the the United States. It is registered in the United States, but does not maintain any IP addresses in the country. It hosts its IPv4 equally amongst the Netherlands, France, Singapore, Brazil, Switzerland, and Germany at 16.7%. For IPv6 addresses, the same countries are listed, but 48.3% of the share is associated with the Netherlands, 48.2% with France, 3.1% with Singapore, and the remaining countries are associated with less than 1%. Link for ASN 57695

AS36236 belongs to NetActuate, Inc, registered with the United States. 68.6% of its IP share is in the United States, and less than 5% are in 20 different countries. The IPv6 share is similar, with the United States at 65.8% South AFrica, Ghana, and Kenya above 9%, and the same IPv4 listed countries for the remaining share. Link for ASN 36236

AS197540 belongs to netcup GmbH, registered in Germany. Germany dominantes its IPv4 share with 89.2%, Austria represents 9.8%, and the Netherlands the remaining 1%. The same three countries constitute the IPv6 share. Link for ASN 197540

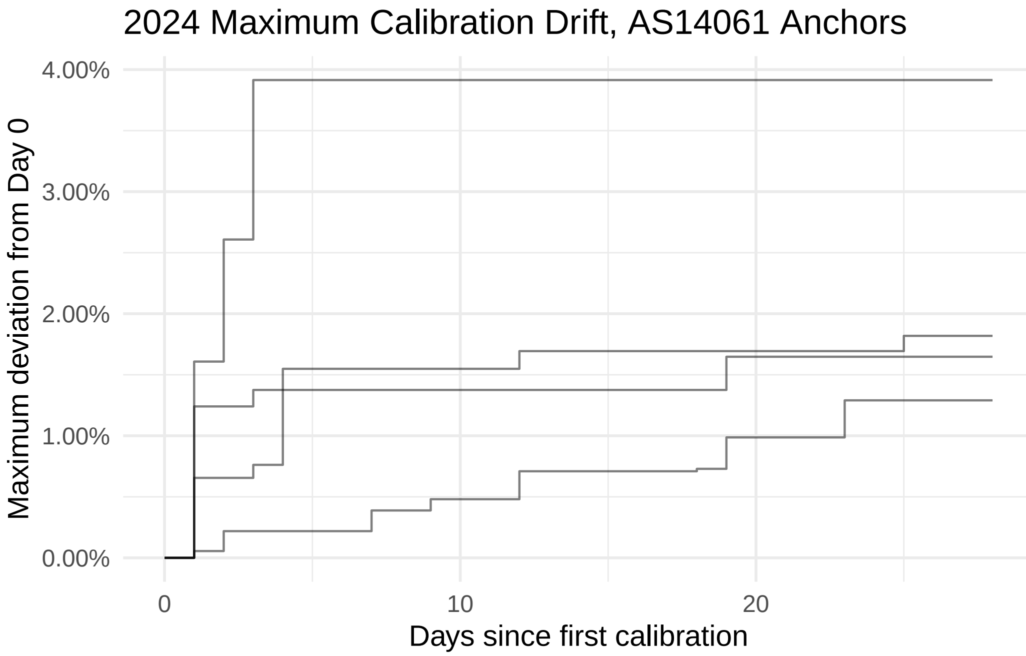
Table 3: Country Frequency

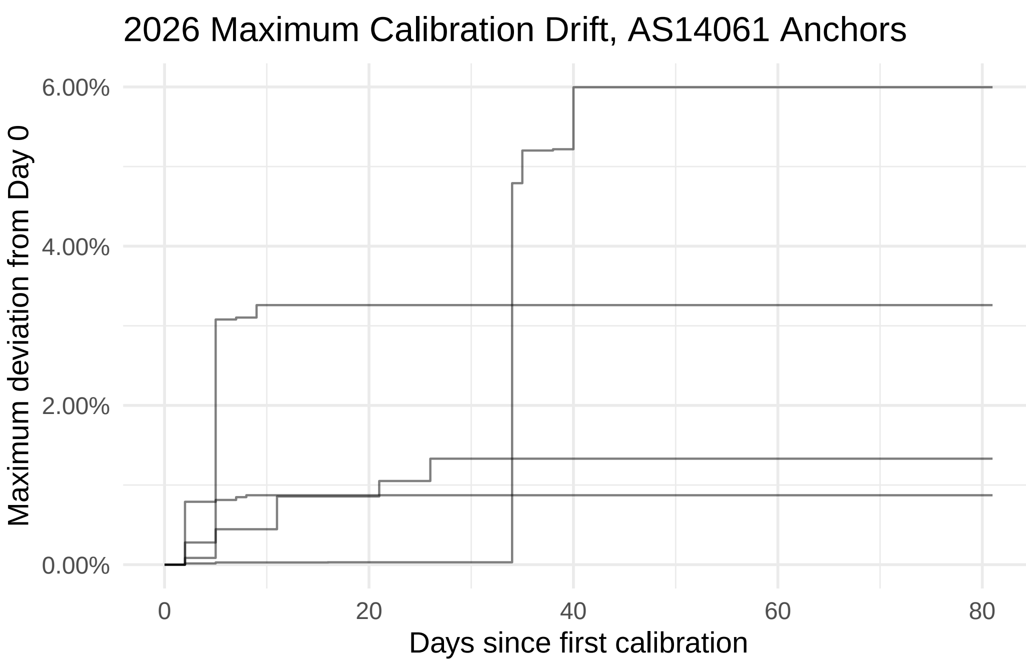
country	anchors
DE	16
US	14
SG	8
AT	7
FR	6
IT	6
CZ	5
GB	5
NL	3

This table lists Germany, the United States, Singapore, Austria, France, Italy, Czechia, United Kingdom, and the Netherlands as being associated with more than 2 anchors that occupy the 90-99.9th percentile consistently. Given the IPv4 and IPv6 shares of the previously identified host network providers, this distribution of countries seems reasonable. The city and continent distributions are available in the code as well, but are omitted.

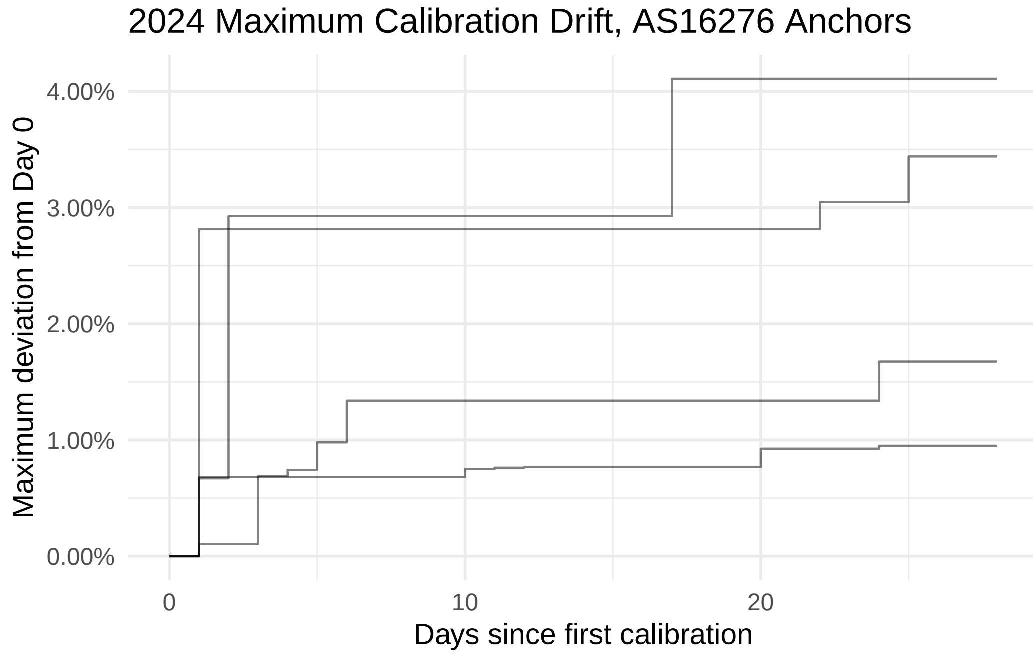
3.3 Visualizing Repeated ASN in 90-99.9th Percentile Range

3.3.1 AS14061

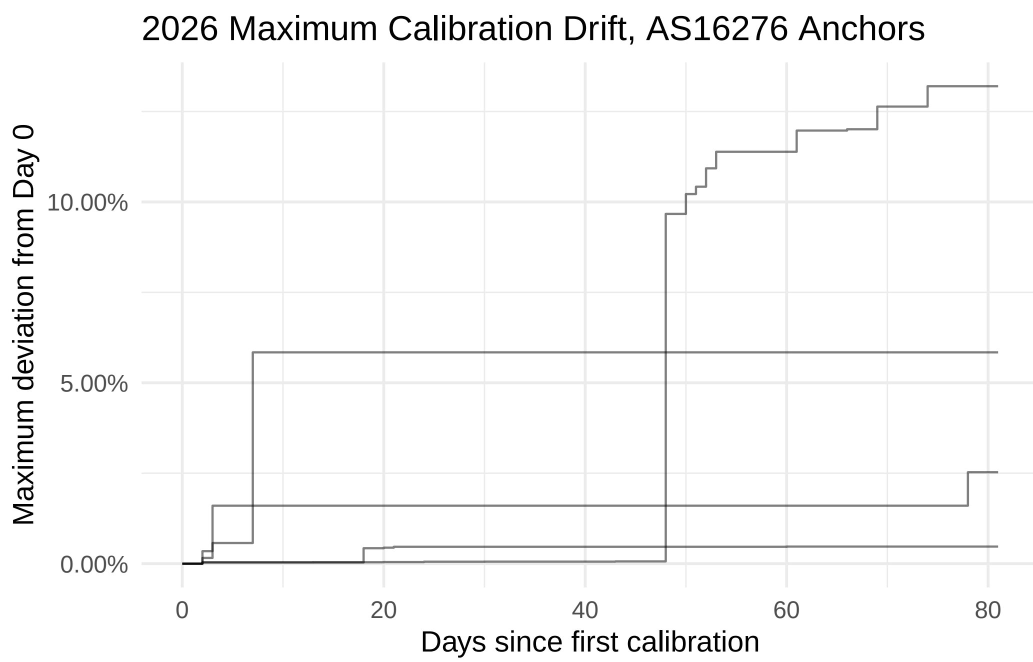




3.3.2 AS16276

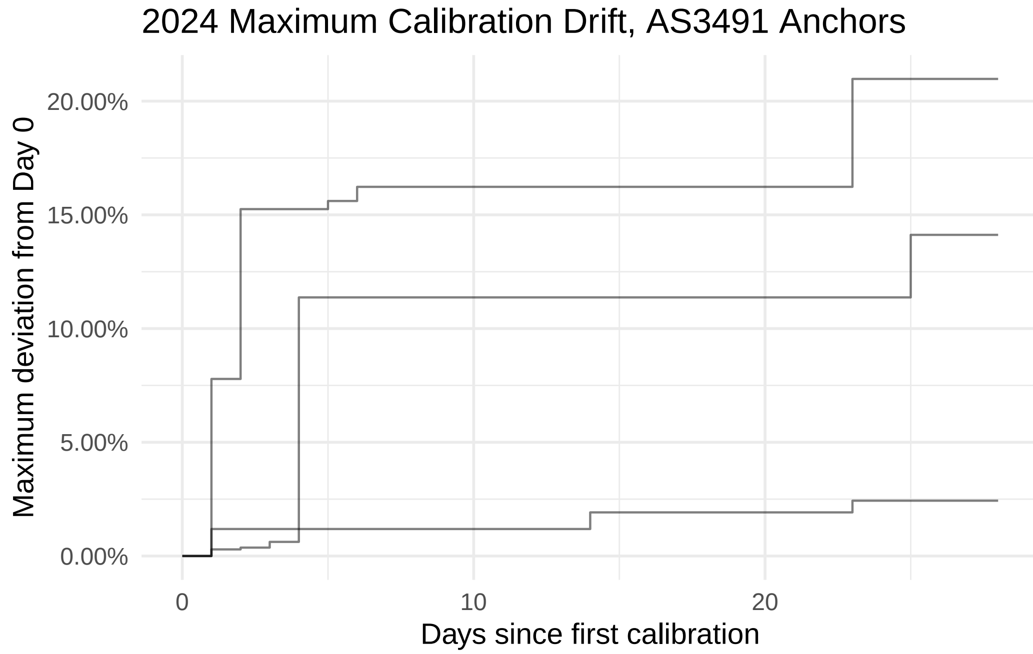


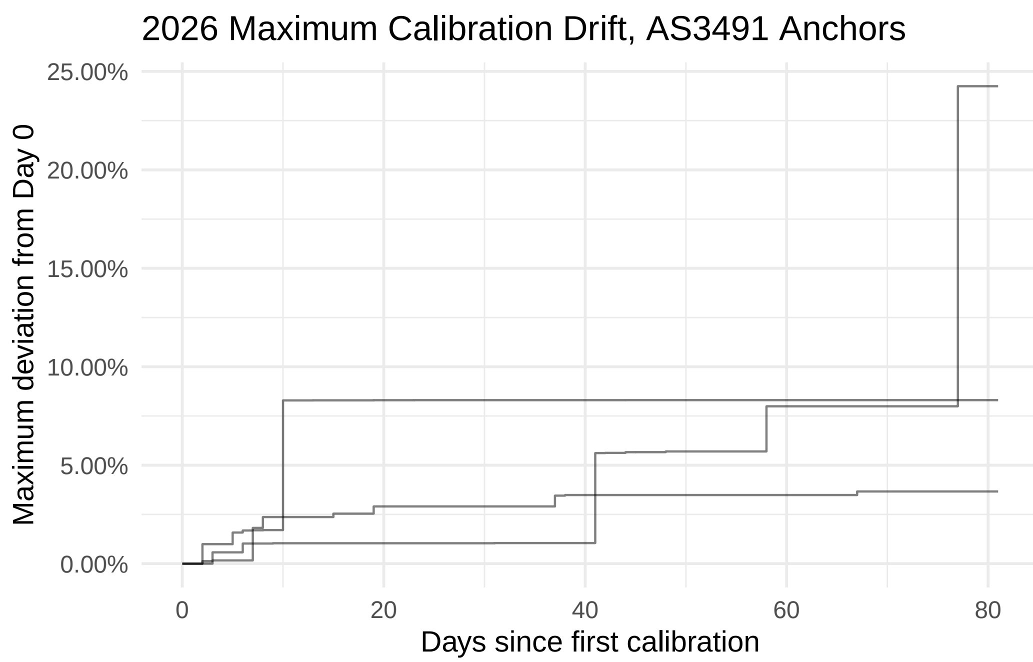
2024 Maximum Calibration Drift, AS16276 Anchors
Maximum deviation from Day 0
4.00%
3.00%
2.00%
1.00%
0.00%
0
10
20
Days since first calibration



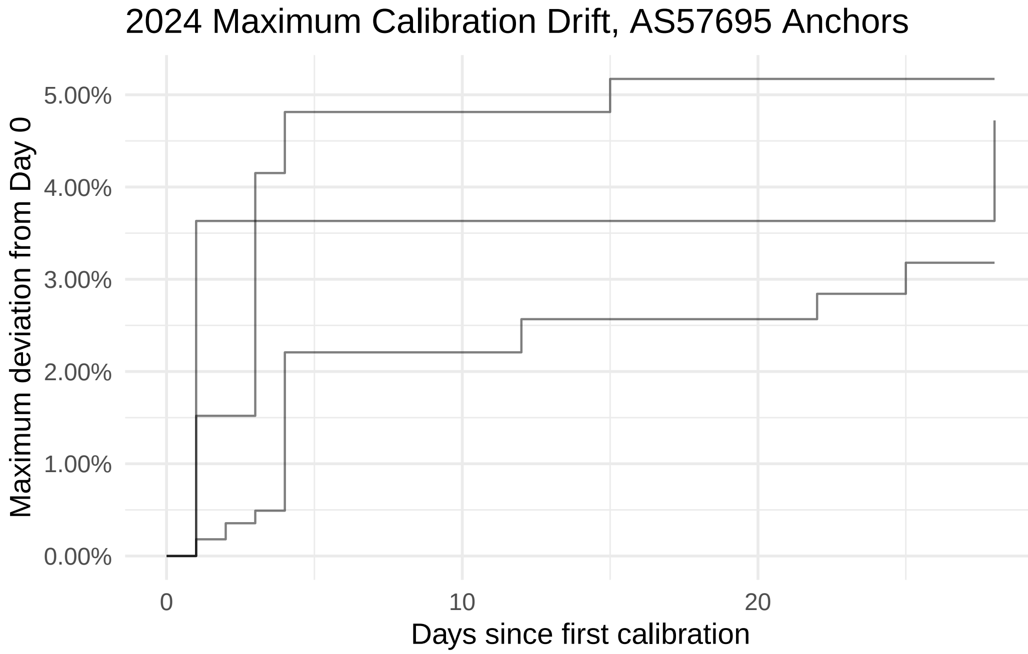
2026 Maximum Calibration Drift, AS16276 Anchors
Maximum deviation from Day 0
10.00%
5.00%
0.00%
0
20
40
60
80
Days since first calibration

3.3.3 AS3491

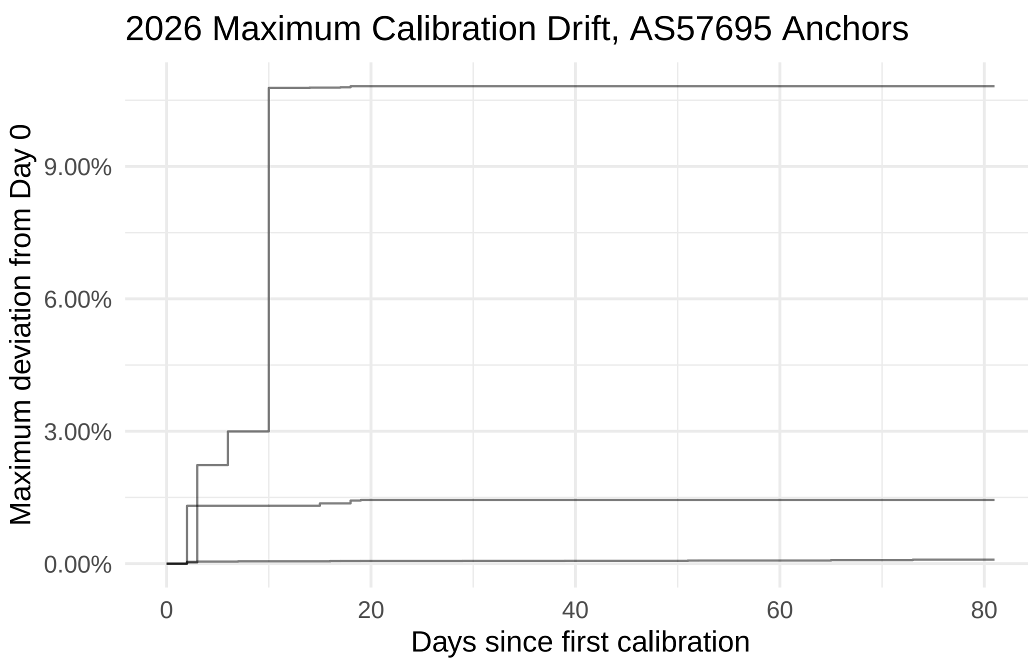

2024 Maximum Calibration Drift, AS3491 Anchors
Maximum deviation from Day 0
20.00%
15.00%
10.00%
5.00%
0.00%
0
10
20
Days since first calibration


2026 Maximum Calibration Drift, AS3491 Anchors
Maximum deviation from Day 0
25.00%
20.00%
15.00%
10.00%
5.00%
0.00%
0
20
40
60
80
Days since first calibration

3.3.4 AS57695

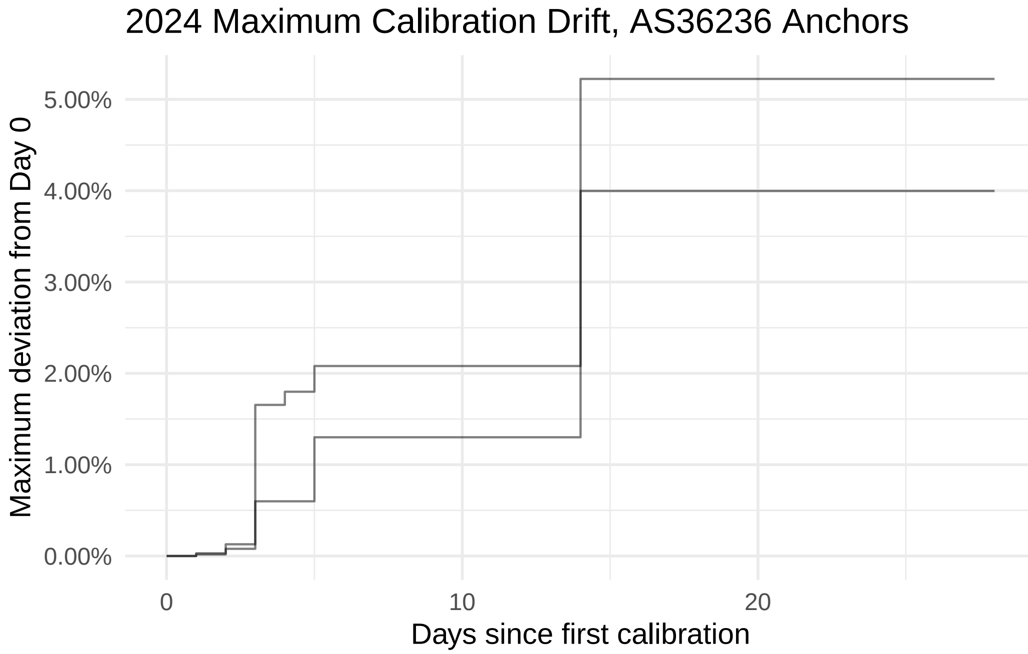


2024 Maximum Calibration Drift, AS57695 Anchors
Maximum deviation from Day 0
5.00%
4.00%
3.00%
2.00%
1.00%
0.00%
0
10
20
Days since first calibration

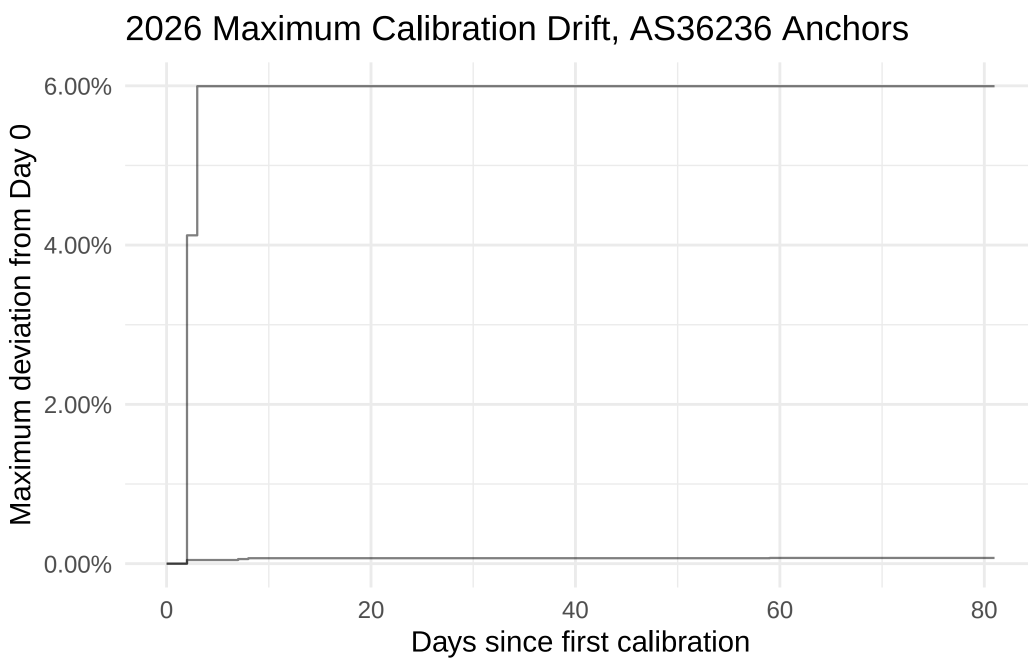


2026 Maximum Calibration Drift, AS57695 Anchors
Maximum deviation from Day 0
9.00%
6.00%
3.00%
0.00%
0
20
40
60
80
Days since first calibration

3.3.5 AS36236

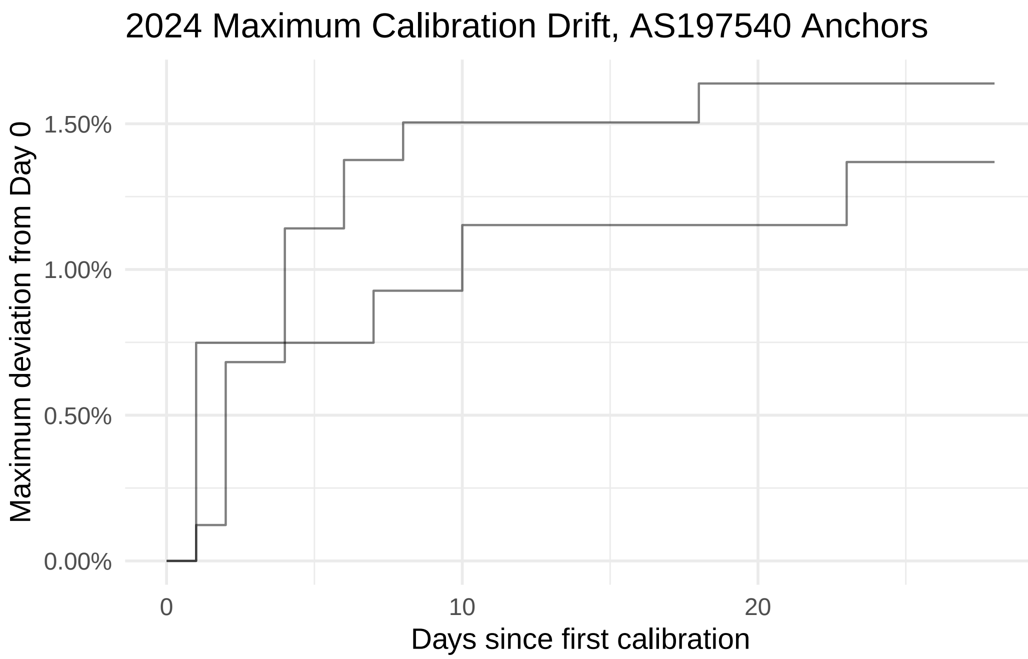


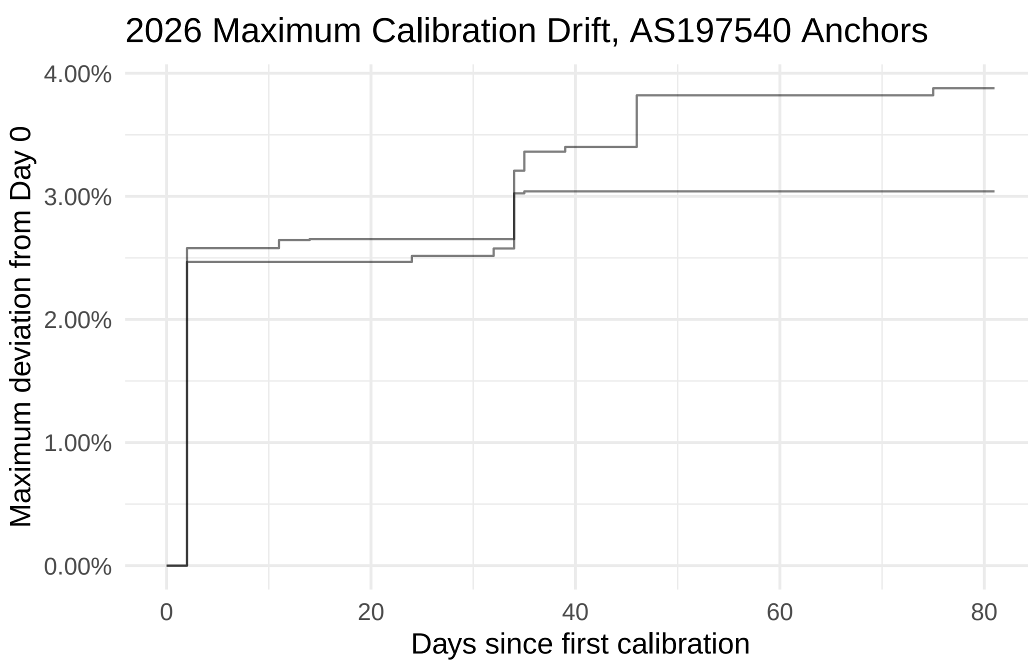
2024 Maximum Calibration Drift, AS36236 Anchors
Maximum deviation from Day 0
5.00%
4.00%
3.00%
2.00%
1.00%
0.00%
0
10
20
Days since first calibration



2026 Maximum Calibration Drift, AS36236 Anchors
Maximum deviation from Day 0
6.00%
4.00%
2.00%
0.00%
0
20
40
60
80
Days since first calibration

3.3.5 AS197540





3.4 Flaw with Maximum Calibration Drift Visualization: Repeated ASN Behavior and Day 0 as Ground Truth

Once we visualize the individual anchor behaviors associated with repeated ASNs withi the 90-999 range, we find that the 5 of 6 ASNs maintain maximum drift below 10% with the exception of one anchor in AS16276 that exceeds 10% in 2026. These 5 ASNs that appeared more than once in the 90-999 range seemed to be associated with anchor IDs with erratic behavior, but appear to be rather conservative upon visualizing their drift behavior. AS3491, however, appears to be responsible for shifting the 99.9th percentile towards 25% in 2026, and indeed occupies the 90-999 range comfortably in 2024.

One ID in AS57675 exceeds 10% drift around day 10 in 2026, but maintains this drift for the remainder of the recorded period. This is Anchor ID 7011, and is in the 90-999 range only once out of 29 days in 2024, but appears in the range 79 of 80 days in 2026. This suggests that the day 0 calibration measurement for 2026 was an outlier, and for the 79 out of 80 days, the calibration value returned to its ground truth calibration range, which was around 10% away from the calibration value recorded on day 0. The maximum drift chart is not enough information to gauge outlier IDs to futher filter our optimal set of reliable IDs, as it determines drift strictly with respect to the day 0 measurement.

Take a look at the individual anchors associated with the ASN Frequency Table 2. Highlighted are two IDs that likely entered the 90-99.9th percentile range due to the strict day 0 ground truth assumption of Maximum Drift visualization.

Table 4: AS14061 Anchors

id	in_range_24	total_24	in_range_26	total26	continent	country	city
6968	9	29	1	80	Asia	IN	Bangalore
6967	1	29	15	80	Europe	DE	Frankfurt
6970	1	29	30	80	Asia	SG	Singapore
6971	1	29	3	80	Europe	GB	London

Table 5: AS16276 Anchors

id	in_range_24	total_24	in_range_26	total26	continent	country	city
6727	4	29	37	80	Oceania	AU	Sydney
6800	3	29	1	80	Europe	PL	Ozarów Mazowiecki
6696	1	29	35	80	Europe	FR	Gravelines
6726	1	29	4	80	Asia	SG	Singapore

Table 6: AS3491 Anchors

id	in_range_24	total_24	in_range_26	total26	continent	country	city
7198	27	29	20	80	Asia	AE	Dubai
7078	26	29	76	80	Oceania	AU	Sydney
7010	1	29	32	80	Asia	SG	Singapore

Table 7: AS57695 Anchors

id	in_range_24	total_24	in_range_26	total26	continent	country	city
6927	15	29	4	80	North America	US	Secaucus
6928	8	29	1	80	North America	US	Reston
7011	1	29	79	80	North America	US	Seattle

Table 8: AS36236 Anchors

id	in_range_24	total_24	in_range_26	total26	continent	country	city
6379	5	29	1	80	North America	US	Durham
6408	1	29	45	80	North America	US	Phoenix

Table 9: AS197540 Anchors

id	in_range_24	total_24	in_range_26	total26	continent	country	city
6581	1	29	27	80	Europe	DE	Nuremberg
6799	1	29	32	80	Europe	DE	Nürnberg

3.5 Geographic Context of Single Anchor Associated ASNs in the 90-99.9 Range

Due to the generally conservative nature of the IDs associated with ASNs frequently appearing in the 90-999 range, we must not overlook the ASNs making individual appearances. Of course, we maintain AS3941 as being associated with frequent outlier drift behavior amongst the repeated ASNs. We will first take a look at the IDs that are associated with unique ASNs that entered the 90-999 range.

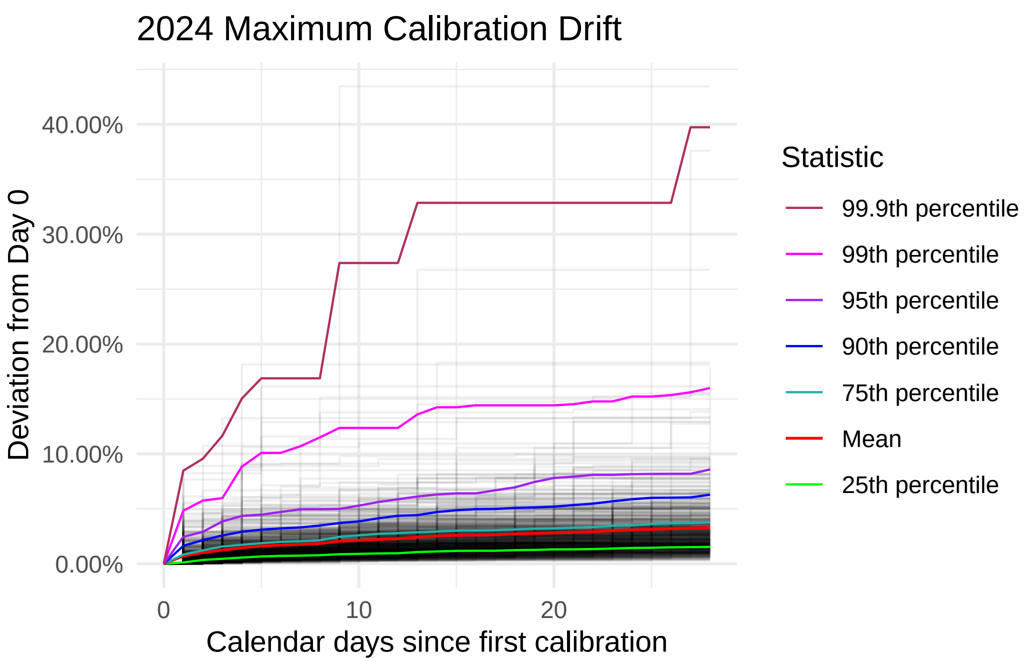
To pare down the IDs for further analysis, we will take away the likely misplaced IDs due to the day 0 ground truth procedure. These IDs will be highlighted in yellow.

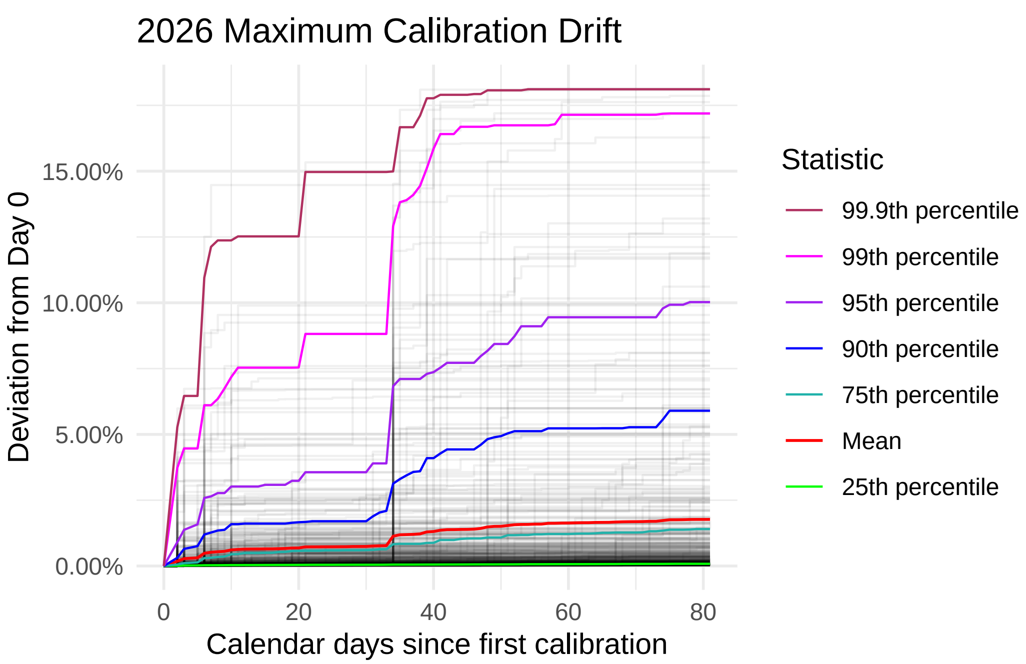
Table 10: Anchors Associated with Unique ASNs

asn	id	in_range_24	total_24	in_range_26	total26	country	city
719	6251	1	29	2	80	FI	Tampere
203329	6313	1	29	76	80	DE	Frankfurt am Main
48943	6325	1	29	49	80	AT	Vienna
2108	6327	1	29	25	80	HR	Zagreb
62416	6328	1	29	73	80	PT	Lisbon
24940	6332	1	29	14	80	DE	Nuremberg
2857	6336	1	29	53	80	DE	Mainz
39878	6360	1	29	49	80	AT	Salzburg
41000	6423	1	29	2	80	GB	Manchester
35574	6435	1	29	49	80	IT	Rozzano
20473	6506	1	29	4	80	SG	Singapore
31034	6509	1	29	49	80	IT	Arezzo
39642	6545	1	29	3	80	DK	Kolding
41095	6568	1	29	4	80	SG	Singapore
39617	6674	1	29	3	80	GB	Balderton
24482	6676	1	29	3	80	SG	Singapore
62310	6751	1	29	2	80	DE	Berlin
41327	6762	1	29	70	80	IT	Palermo
46997	6794	1	29	71	80	US	Seattle
24971	6809	1	29	26	80	CZ	Brno
199399	6841	1	29	44	80	GR	Athens
21320	6874	1	29	70	80	FR	Paris
62513	6888	1	29	4	80	CA	Toronto
38064	6904	1	29	5	80	NZ	Auckland
39835	6922	1	29	48	80	DE	Göttingen
39636	7003	1	29	36	80	NL	Arnhem
30971	7008	1	29	42	80	AT	Vienna
13101	7012	1	29	8	80	DE	Kiel
38001	7091	1	29	2	80	SG	Singapore
56381	7097	1	29	79	80	DE	Frankfurt am Main
21034	7110	1	29	2	80	IT	Pescara
33920	7125	1	29	5	80	GB	Leeds
29423	7141	1	29	31	80	DE	Frankfurt
50030	7156	1	29	28	80	DE	Berlin

39063	7169	1	29	49	80	DE	Frankfurt
59678	7193	1	29	33	80	US	Piscataway
60841	7201	1	29	4	80	NL	Amsterdam
25248	7211	1	29	16	80	CZ	Prague
35492	7216	1	29	49	80	AT	Vienna
54316	7224	1	29	2	80	US	Greenwich
29287	7225	1	29	49	80	AT	Vienna
23428	7238	1	29	61	80	US	Kansas City
14907	7261	1	29	60	80	NL	Amsterdam
200187	7265	1	29	49	80	DE	Frankfurt am Main
216444	7272	1	29	28	80	SG	Singapore
32167	7292	1	29	2	80	HK	Hong Kong
36444	6389	2	29	1	80	US	Detroit
48362	6476	2	29	1	80	AT	Feldkirch
204092	6494	2	29	1	80	FR	Rennes
202297	6547	2	29	1	80	CZ	Prague
199188	6559	2	29	1	80	GB	London
395823	7043	2	29	78	80	US	Seattle
31510	7083	2	29	1	80	AT	Innsbruck
8473	7127	2	29	1	80	SE	Malmö
24840	7196	2	29	49	80	DE	Frankfurt
60396	6441	3	29	1	80	SE	Lund
59715	6496	3	29	1	80	IT	San Benedetto del Tronto
199610	6649	3	29	32	80	US	Los Angeles
29169	6363	4	29	1	80	FR	Saint-Denis
201333	6675	5	29	1	80	IT	Piacenza
15699	7086	6	29	2	80	ES	Alcalá de Henares
47295	7096	6	29	1	80	DE	Bautzen
25192	7223	6	29	1	80	CZ	Prague
7752	7221	8	29	79	80	US	Seattle
6881	7024	9	29	1	80	CZ	Prague
209097	6736	12	29	1	80	FR	Trevoux
53698	7159	14	29	48	80	US	Phoenix
13303	7213	15	29	1	80	RS	Nis
12008	6954	19	29	36	80	IN	Mumbai
16347	6443	22	29	1	80	FR	Maxeville
15802	6962	26	29	1	80	AE	Dubai
202196	6639	28	29	1	80	HK	Hong Kong
206186	6323	29	29	1	80	PL	Katowice

3.6 Filtered Maximum Drift (After Day 0 Adjustment)





3.7 Discussion on Adjusted Maximum Drift Visualizations

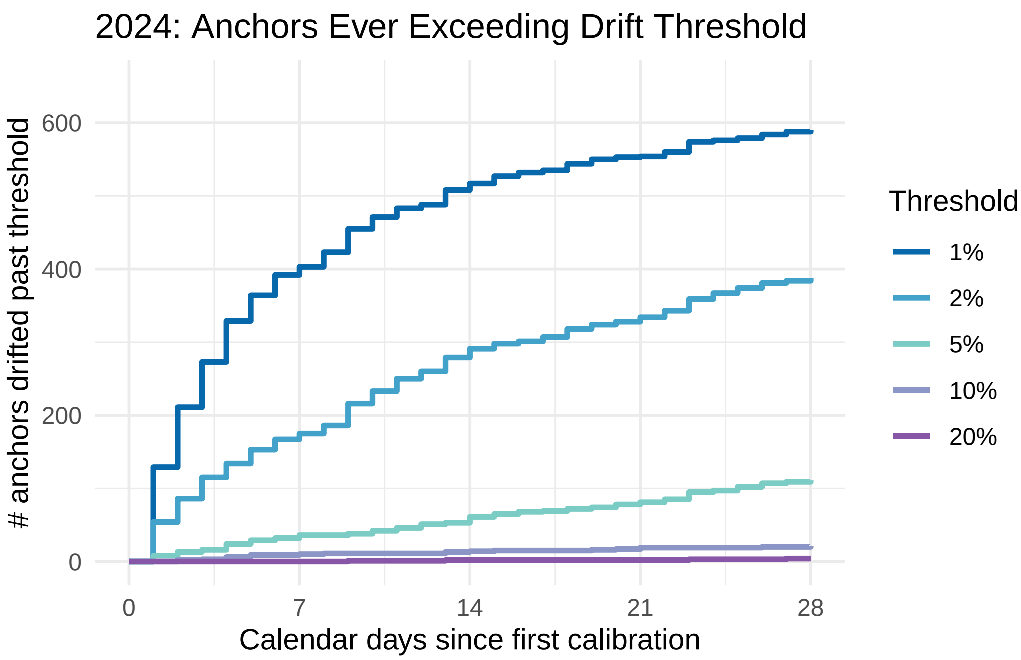
Upon identifying which anchors appeared the most in the 90-99.9th percentile range, we found that 17 of 91 anchors spent nearly all days of the recorded period in the range for one year, while only appearing the range once the other year. Furthermore, Anchor 7078 recorded 90-99.9th percentile drift 26 of 29 days in 2024 and 76 of 80 days in 2026. As the maximum calibration drift plot evaluates percentage of drift strictly with respect to the day 0 measurement, if anchors are to be at an outlier calibration value on the first day of the recorded period, a prominent drift would likely be maintained for the rest of the period as the calibration returns to its usual value. This suggests that the initial calibration values of 18 anchors including Anchor 7078 were misrepresented on day 0, and brought back to their average rates when appearing in the 90-99.9th percentile range.

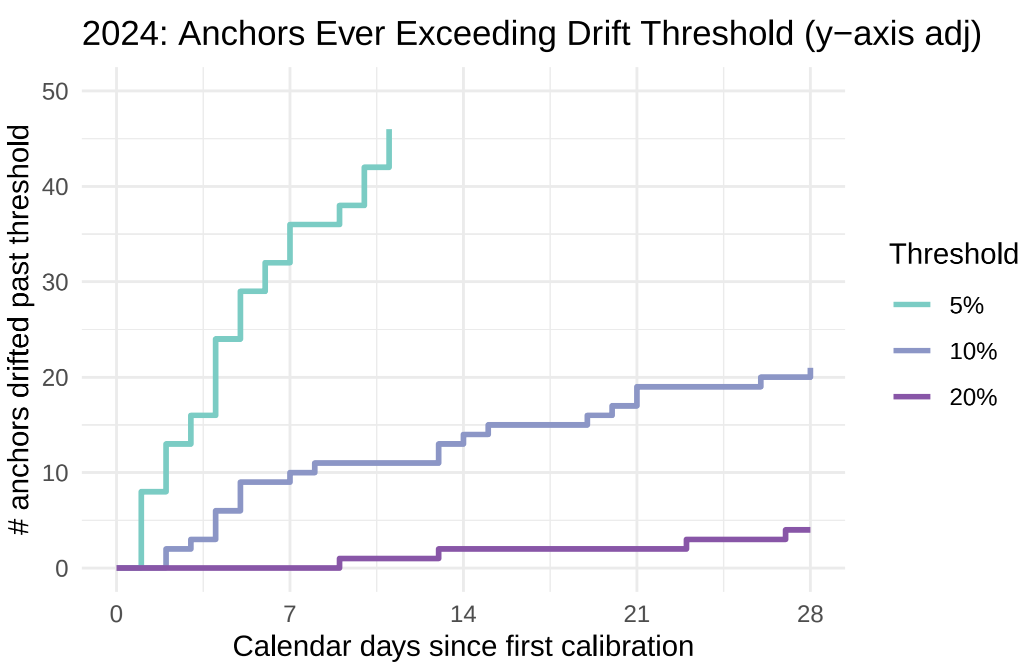
We control for this flaw by removing the anchors that appear in the range more than 75% of the time in one year, while not appearing very few times in the other, or if the anchor appears in range for more than 75% of the days for both years. Ideally, we are then left with a more accurate representation of the erratic anchors whose calibration values drift significantly from day to day.

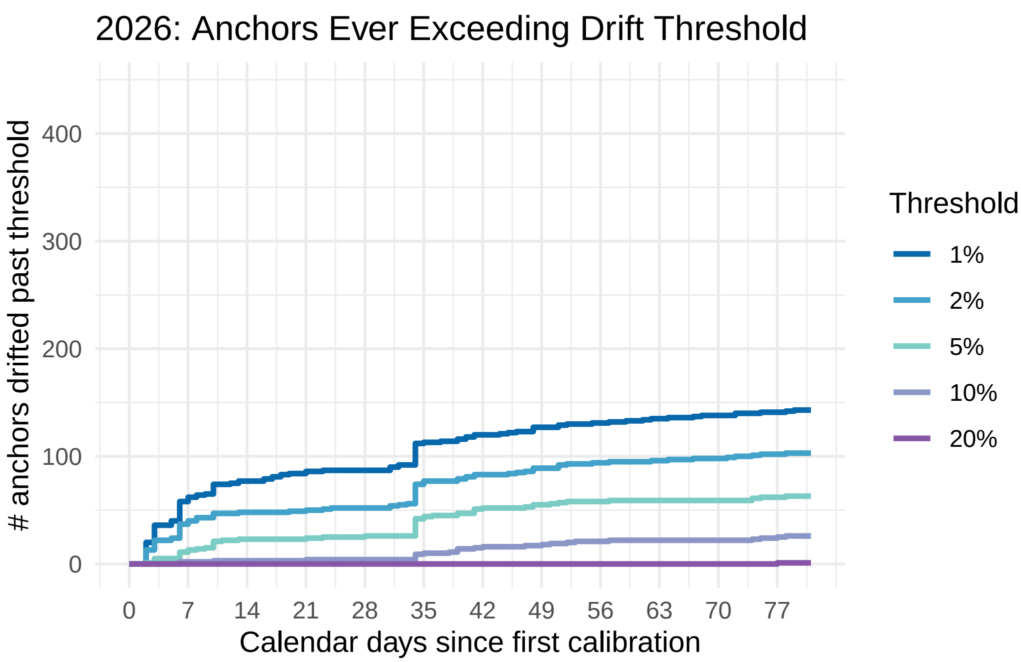
After removing the aforementioned 17 anchors, we plot maximum drift once again in 3.5.1. As the majority of the anchors removed recorded their majority 90-99.9th percentile range days in 2026 and spent 1 to at most 8 days in range in 2024, the adjustment from the original 2026 drift from 0-25% to 0-18% is indeed more prominent than the 2024 plot that maintained its 0-45% drift. Indeed, this is a very granular process, and we benefit from considering other visualizations to better understand anchor behavior.

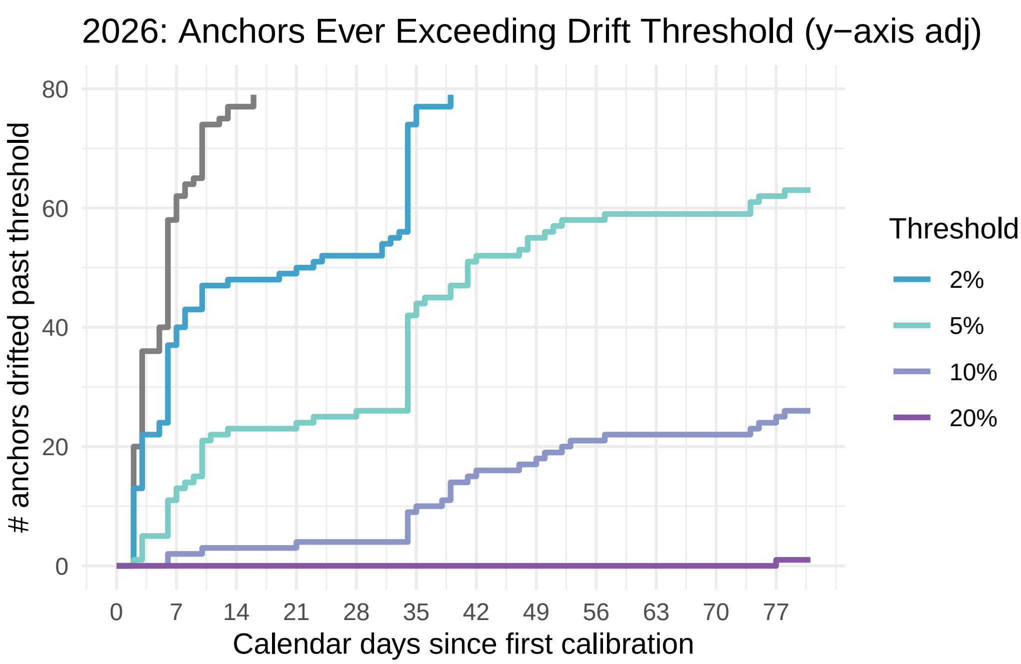
Using Zach's 2024 analysis as a reference once again, the Maximum Drift Threshold Visualization offers another perspective in understanding our data. The following visualizations retain all anchor IDs and ASNs that were removed in this section, and represents the same data as the Maximum Drift Visualizations in *Section 3. Maximum Drift.*

4. Maximum Drift Threshold






2026: Anchors Ever Exceeding Drift Threshold
anchors drifted past threshold
400
300
200
100
0
0
7
14
21
28
35
42
49
56
63
70
77
Calendar days since first calibration
Threshold
1%
2%
5%
10%
20%


2026: Anchors Ever Exceeding Drift Threshold (y−axis adj)
anchors drifted past threshold
80
60
40
20
0
0
7
14
21
28
35
42
49
56
63
70
77
Calendar days since first calibration
Threshold
2%
5%
10%
20%

4.1 Observations/Discussion

The Maximum Drift Threshold uses percentage thresholds to illustrate the rate at which the anchors exceed 1%, 2% 5%, 10%, and 20% drift. By day 7, nearly 400 of 659 anchors in the 2024 data have exceeded 1% drift, and around 200 exceed 2% and 55 exceed 5% by day 9 and 14 respectively. In order to improve interpretability, I have adjusted the y axis for examining the 10% and 20% specific thresholds trends. In this figure, we see that about 20 of 659 anchors exceeded 10% drift, and 4 of 659 exceeded 20% by the end of the recorded period. This visualization is especially convenient for observing the rate at which the anchors collectively exceed each threshold.

For the 2026 data, we use two figures to visualize our threshold details carefully once more. The drift trends for the 2026 data appear to be more conservative than the 2024 data, which can be credited to the comparatively wider date range and therefore less consistent calibration retrieval (leaving us with 444 of 899 anchors). Nonetheless, this shift that we have been noticing around day 35 persists, this time quantifiable as around 30 new anchors reaching 2%, and 20 additional anchors reaching 5% drift thresholds. It appears that a total of around 25 anchors reach the 10% threshold by day 77, and only 1 anchor exceeds 20% over the entire recorded period.

It is encouraging to observe that in the 2024 data, the number exceeding the 20% threshold does not exceed 5 anchors, and in 2026, it is merely 1. Our previous plot appeared failed to visually quantify the number of anchors accumulating the most drift, and to understand the ratio of anchors drifting past a 20% threshold as 4 of 659 anchors is encouraging in grasping the proportion of “erratic” or “volatile” anchors to “stable” anchors that maintain 1-5% drift.

4.2 Anchor IDs that Exceeded 20%, 10%, and 5%

4 Anchor IDs Exceeded 20%: 7192, 7198, 7207, 7222

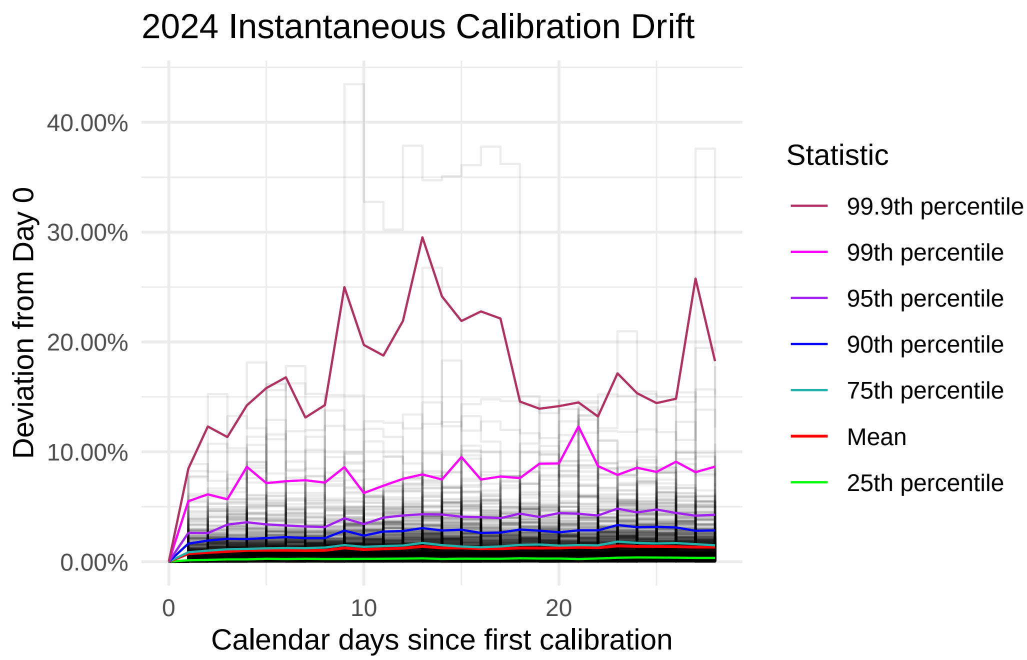
21 Anchor IDs Exceeded 10%: *6392, 6398, 6410, 6586, 6718, 6782, 6882, 6887, 6935, 6977, 6994, 7050, 7077,* 7078, **7192**, **7198**, **7207**, *7219*, **7222**, *7239, 7251*

1 Anchor ID Exceeded 20%: 7198

26 Anchor IDs Exceeded 10%: 6313, 6325, 6360, 6509, *6696*, 6809, 6841, 6922, *6954*, 7003, 7008, 7011, 7097, *7128*, 7169, *7181*, **7198**, 7216, 7225, *7456*, *7502, 7511, 7574, 7616, 7635, 7680*

We previously discussed anchor ID 7198 in Table 6, concerning anchors associated with AS3491. The anchor occupied the 90-99.9th percentile range for 27 of 29 days in 2024, and 20 of 80 days in 2026. The other IDs 7192, 7207, and 7222 that exceeded 20% in 2024 and do not appear in the ID summaries from *Section 3.4* and *Section 3.5.* The italicized anchors are all omitted from the *Section 3.4* and *Section 3.5* summaries, and provide new data for us to consider in terms of a new subset of outlier anchors.

5. Instantaneous Drift Visualization



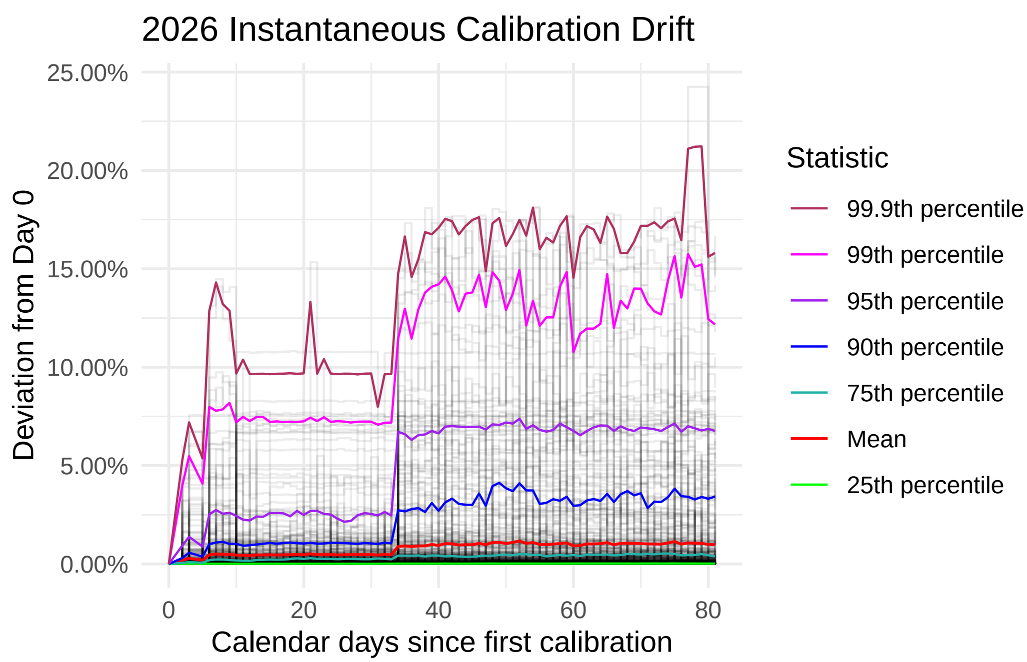
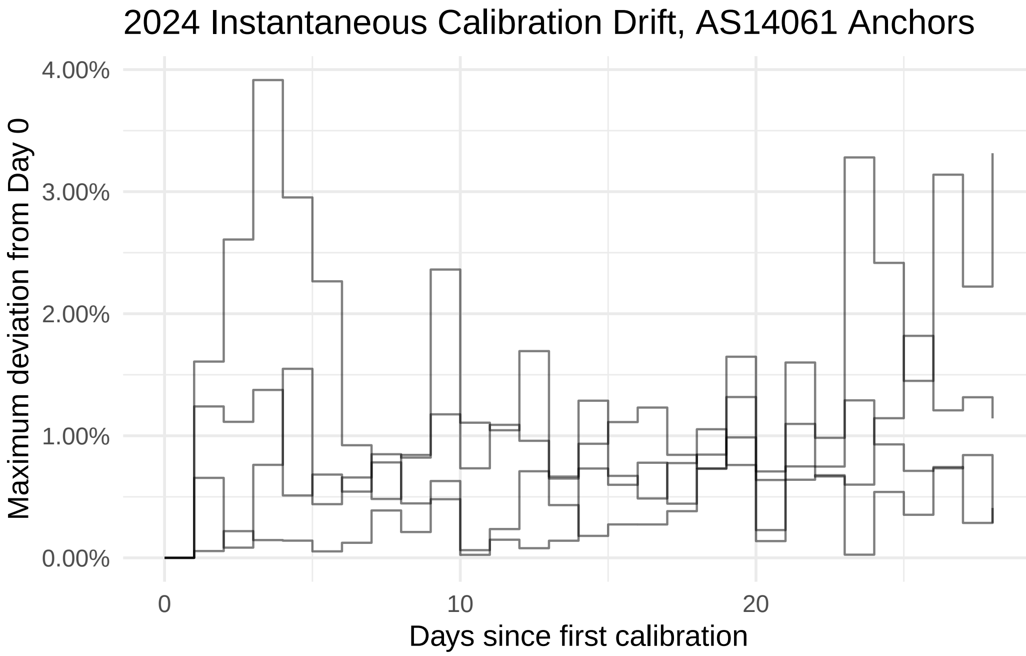


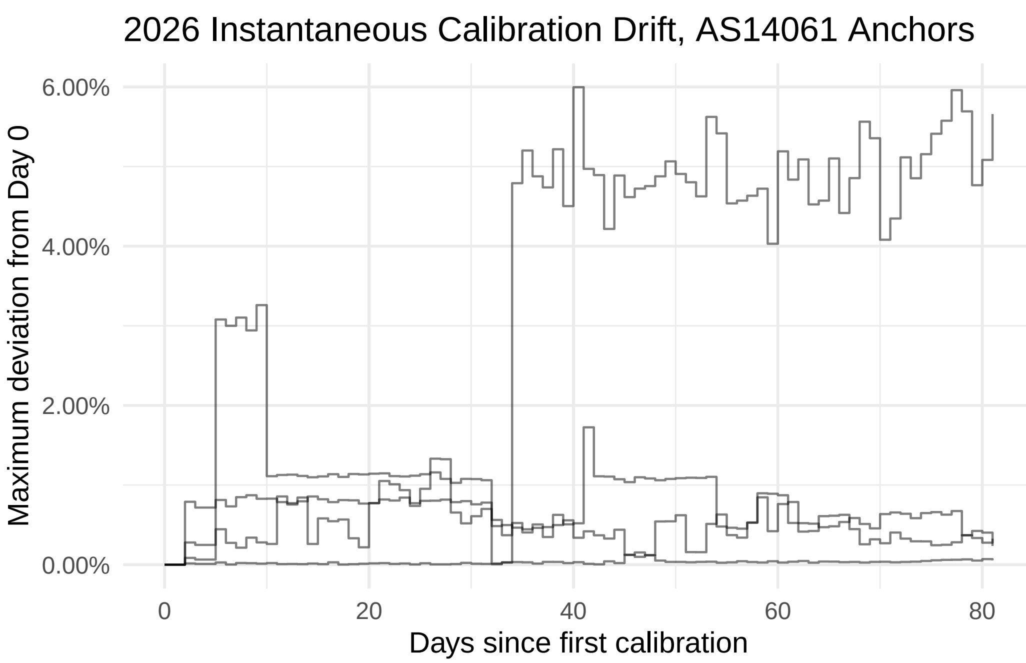
Table 11: ASN Frequency

asn	anchors
3491	4
14061	4
16276	4
36236	3
57695	3
199524	3
8473	2
16347	2
20473	2
24940	2
35574	2
39923	2
197540	2
199610	2

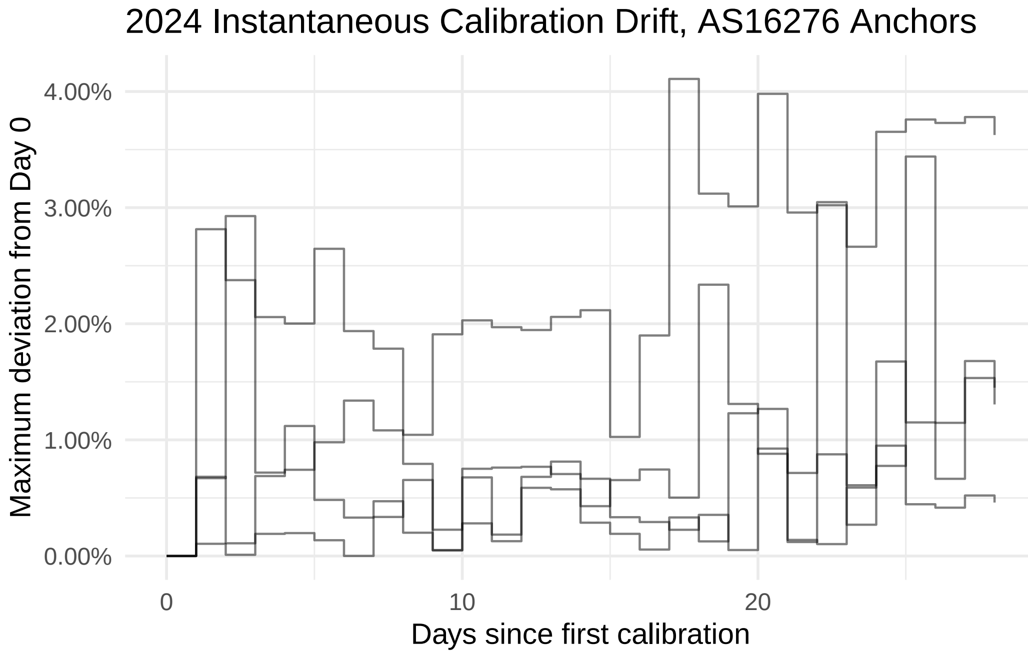
5.3 Visualizing Repeated ASN in 90-99.9th Percentile Range

5.3.1 AS14061

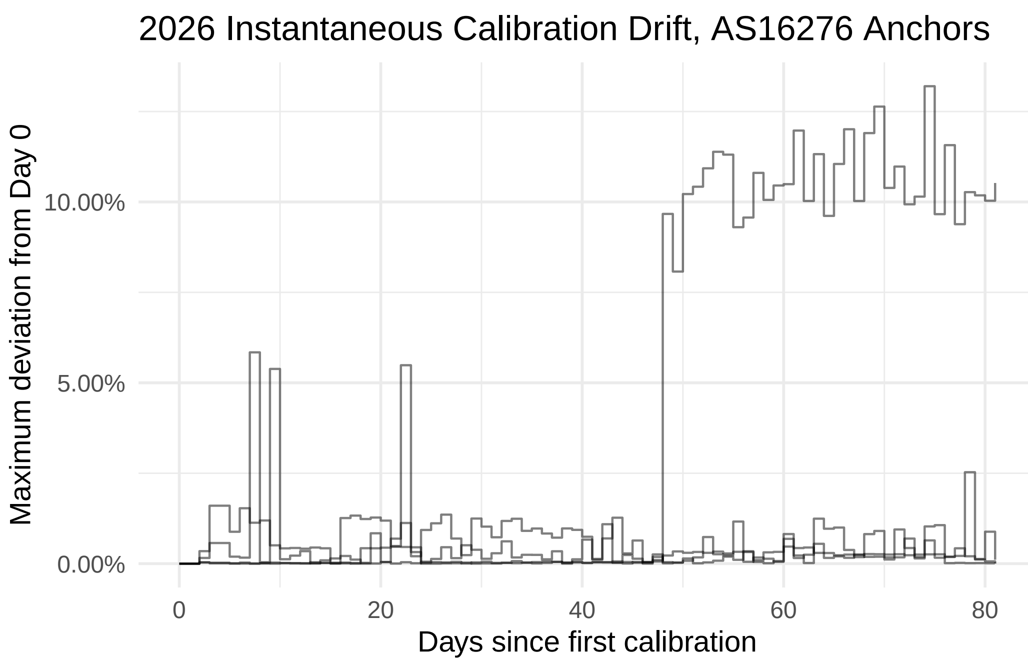




5.3.2 AS16276

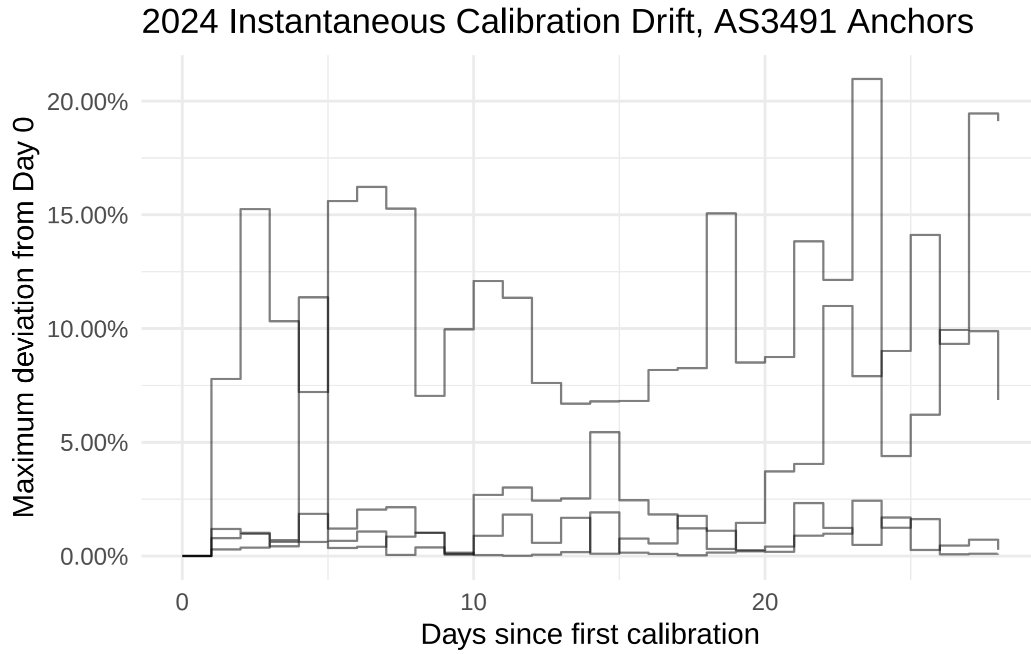


2024 Instantaneous Calibration Drift, AS16276 Anchors
Maximum deviation from Day 0
4.00%
3.00%
2.00%
1.00%
0.00%
0
10
20
Days since first calibration

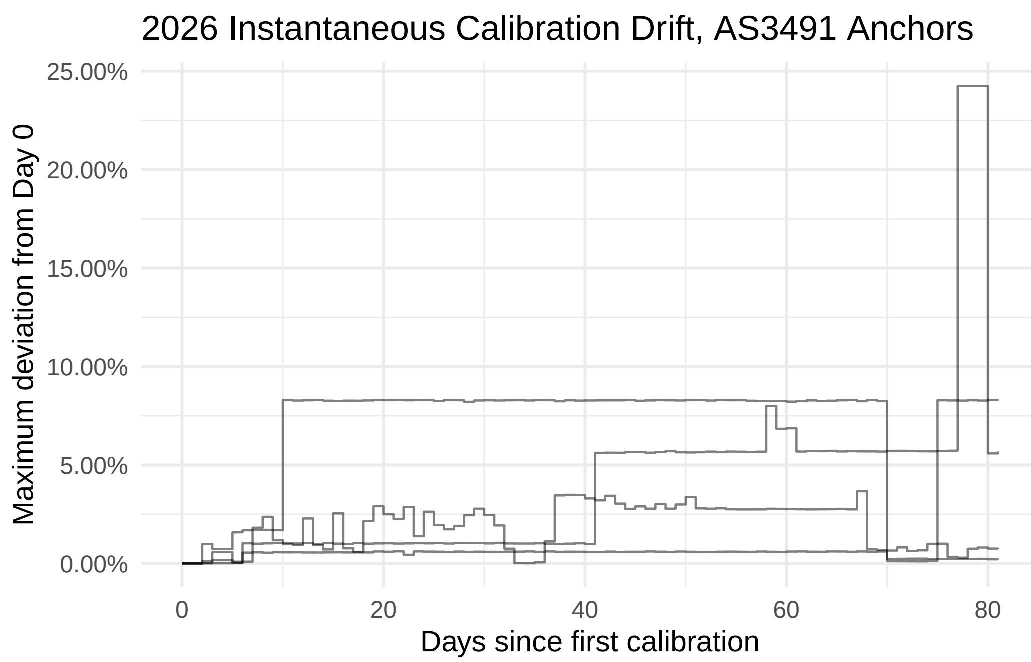


2026 Instantaneous Calibration Drift, AS16276 Anchors
Maximum deviation from Day 0
10.00%
5.00%
0.00%
0
20
40
60
80
Days since first calibration

5.3.3 AS3491

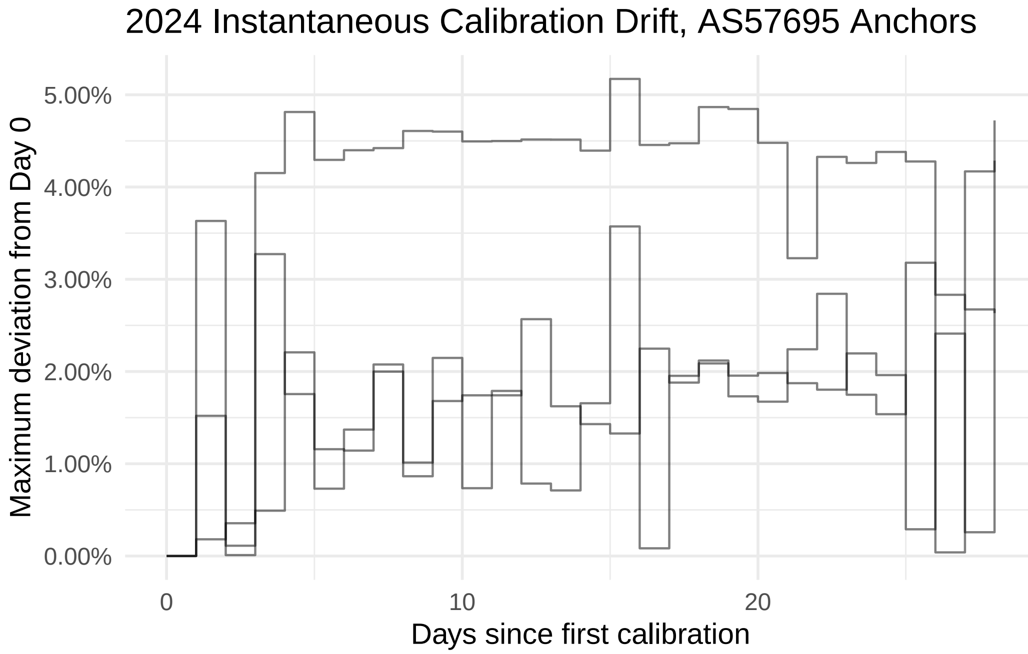


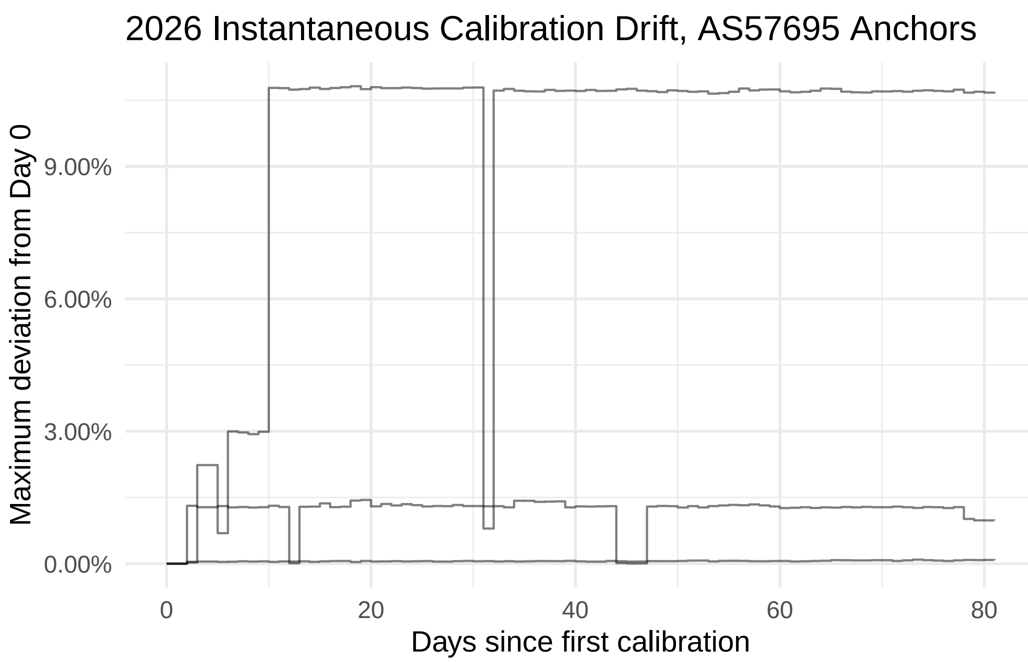
2024 Instantaneous Calibration Drift, AS3491 Anchors
Maximum deviation from Day 0
20.00%
15.00%
10.00%
5.00%
0.00%
0
10
20
Days since first calibration



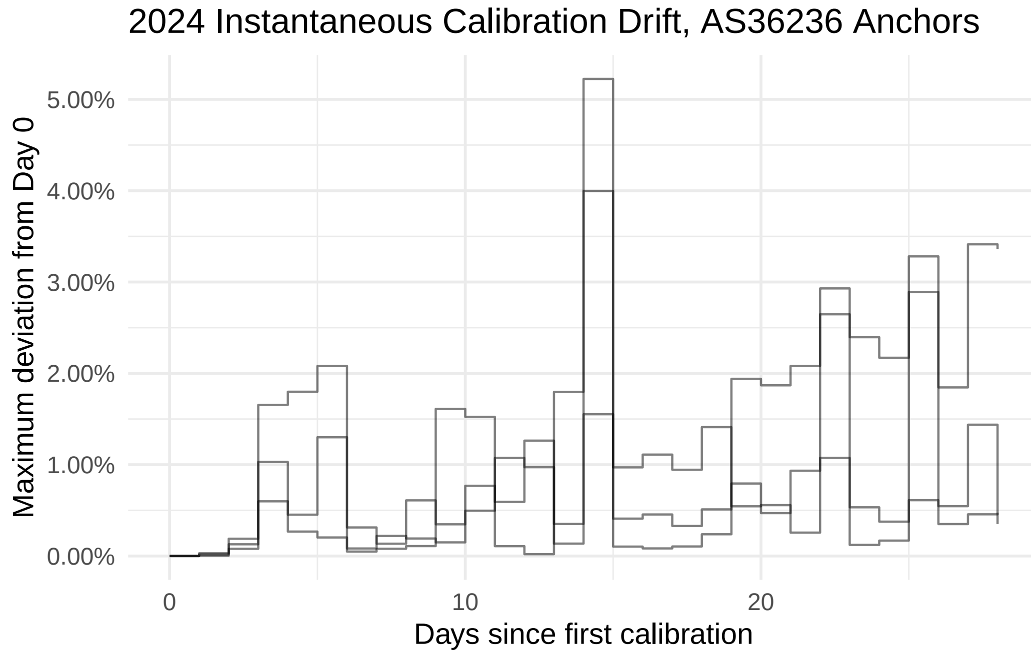
2026 Instantaneous Calibration Drift, AS3491 Anchors
Maximum deviation from Day 0
25.00%
20.00%
15.00%
10.00%
5.00%
0.00%
0
20
40
60
80
Days since first calibration

5.3.4 AS57695

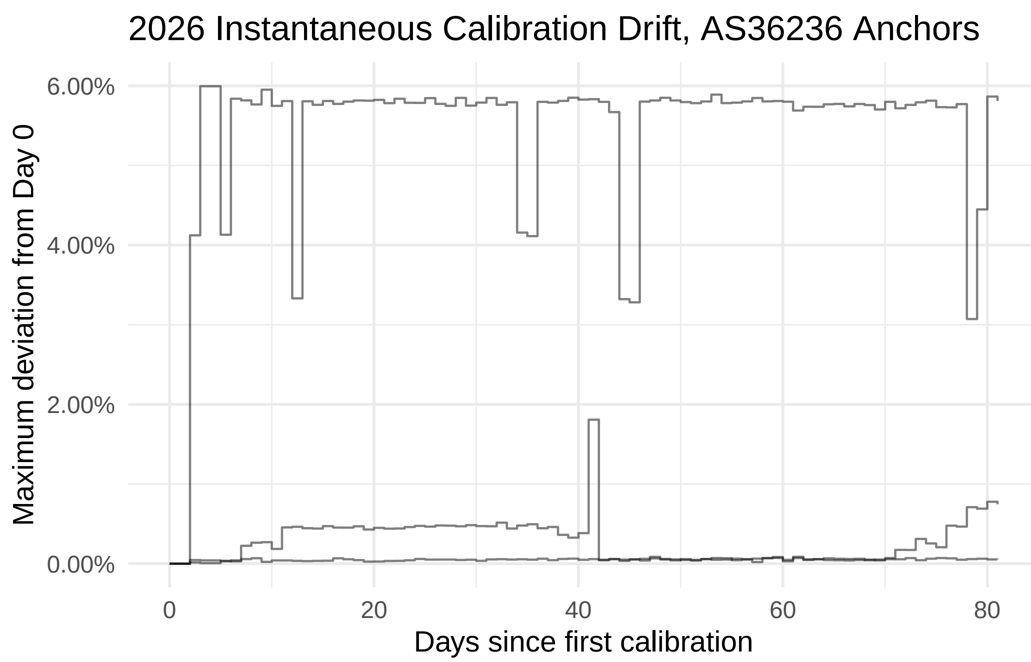

2024 Instantaneous Calibration Drift, AS57695 Anchors
Maximum deviation from Day 0
5.00%
4.00%
3.00%
2.00%
1.00%
0.00%
0
10
20
Days since first calibration


2026 Instantaneous Calibration Drift, AS57695 Anchors
Maximum deviation from Day 0
9.00%
6.00%
3.00%
0.00%
0
20
40
60
80
Days since first calibration

5.3.5 AS36236

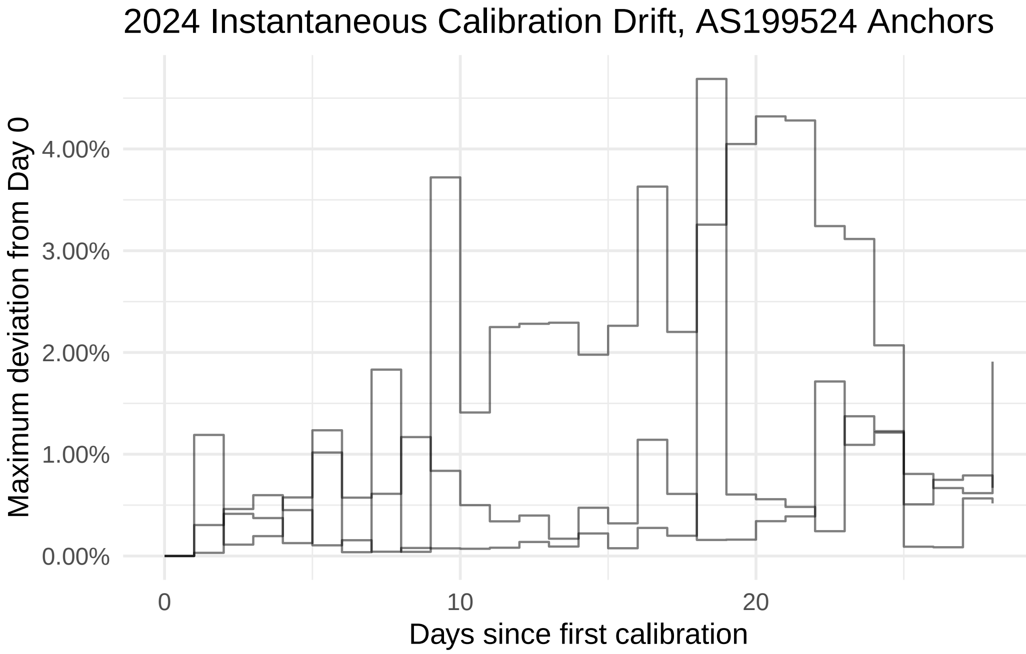


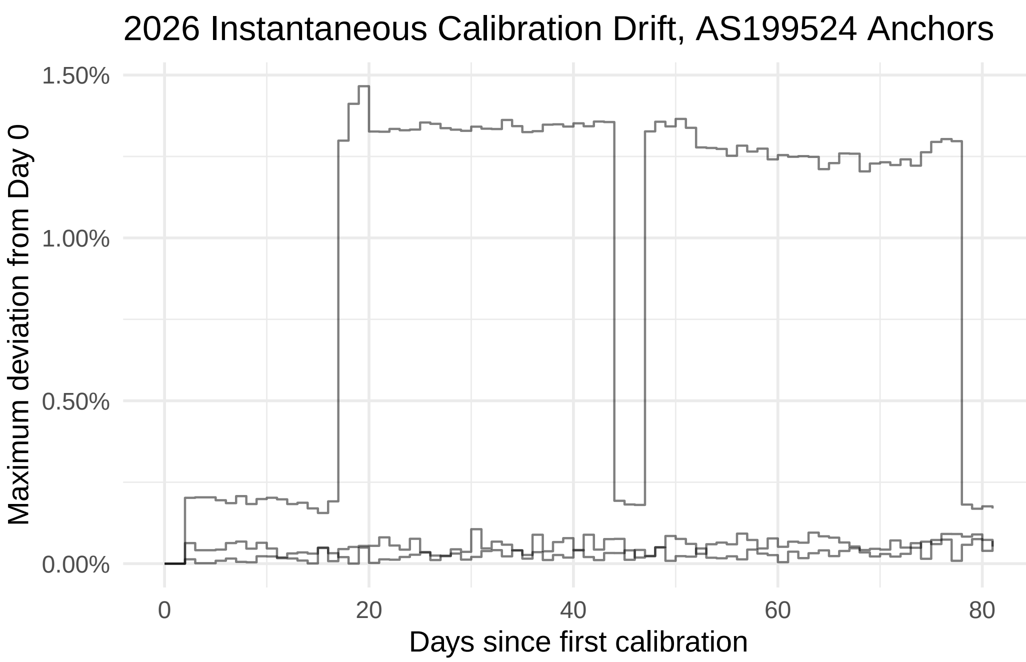
2024 Instantaneous Calibration Drift, AS36236 Anchors
Maximum deviation from Day 0
5.00%
4.00%
3.00%
2.00%
1.00%
0.00%
0
10
20
Days since first calibration



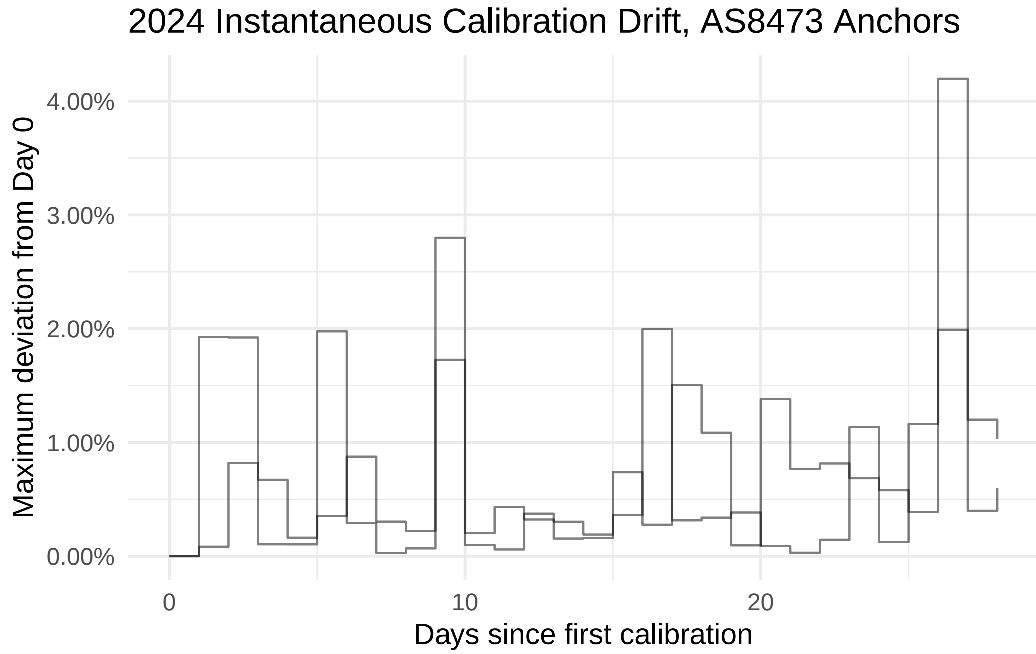
2026 Instantaneous Calibration Drift, AS36236 Anchors
Maximum deviation from Day 0
6.00%
4.00%
2.00%
0.00%
0
20
40
60
80
Days since first calibration

5.3.5 AS199524

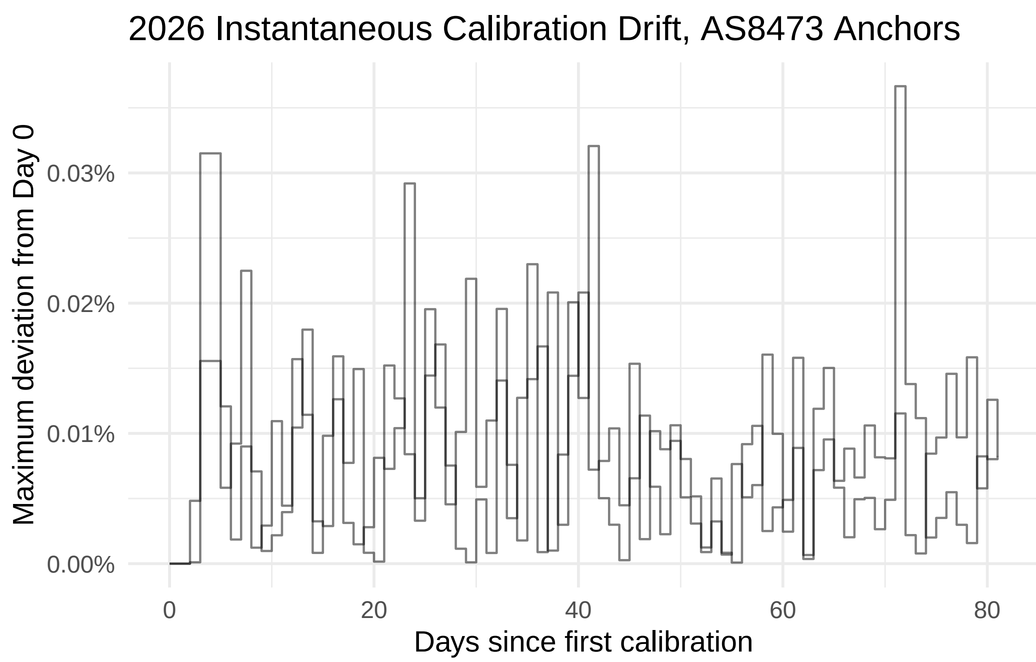




5.3.5 AS8473

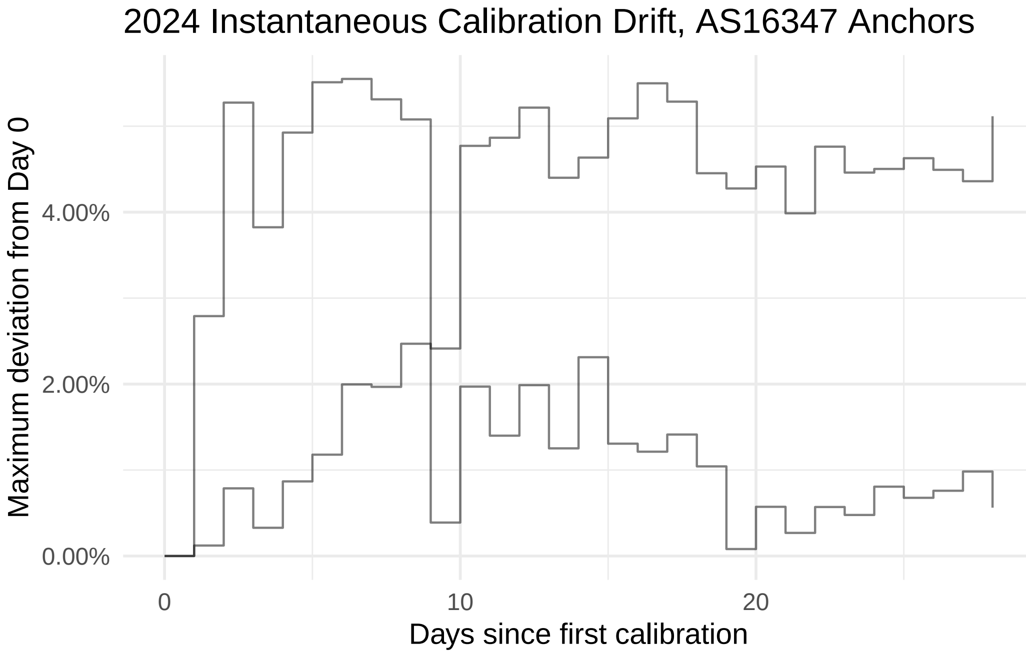


2024 Instantaneous Calibration Drift, AS8473 Anchors
Maximum deviation from Day 0
4.00%
3.00%
2.00%
1.00%
0.00%
0
10
20
Days since first calibration

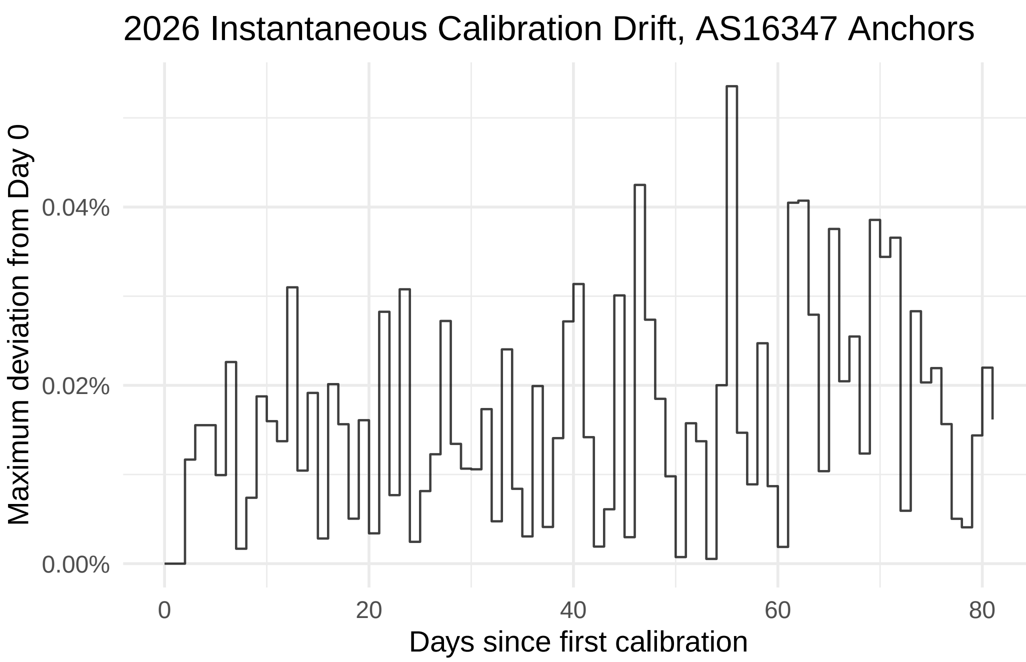


2026 Instantaneous Calibration Drift, AS8473 Anchors
Maximum deviation from Day 0
0.03%
0.02%
0.01%
0.00%
0
20
40
60
80
Days since first calibration

5.3.5 AS16347

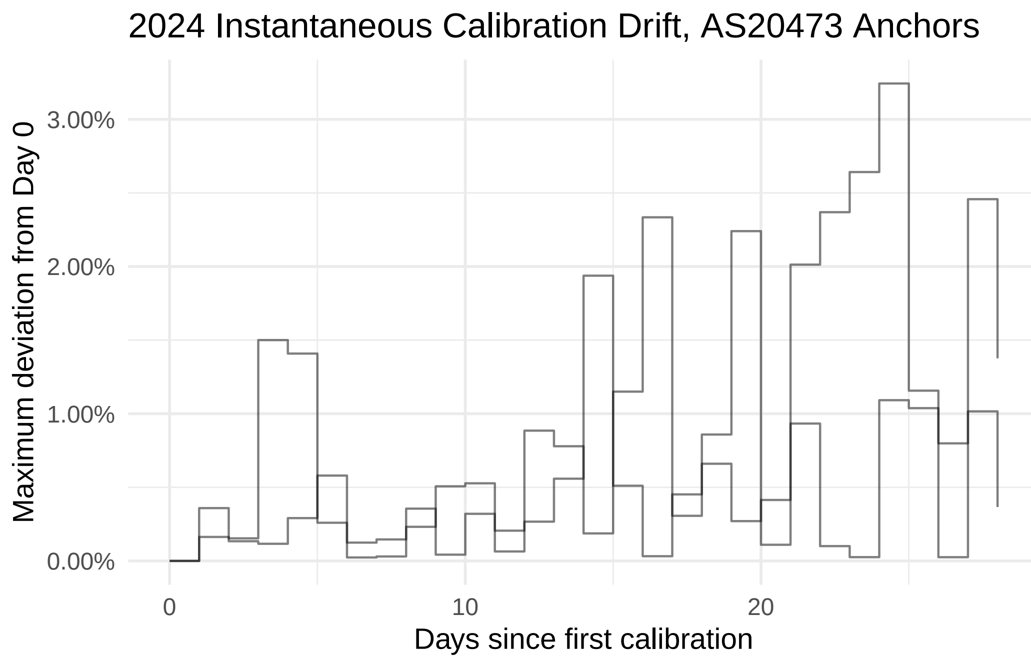


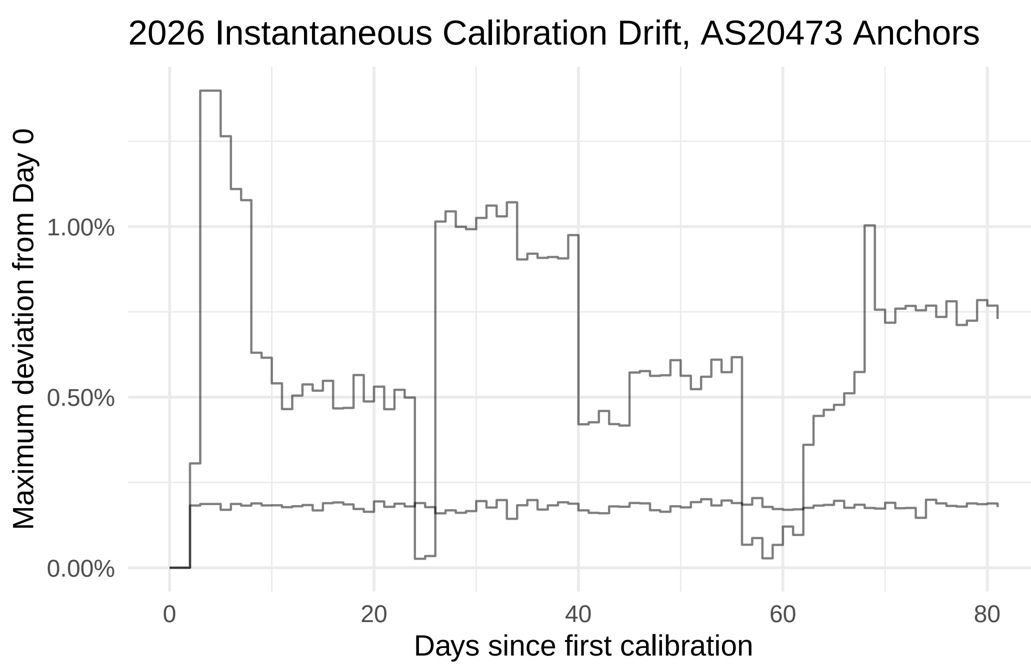
2024 Instantaneous Calibration Drift, AS16347 Anchors
Maximum deviation from Day 0
4.00%
2.00%
0.00%
0
10
20
Days since first calibration



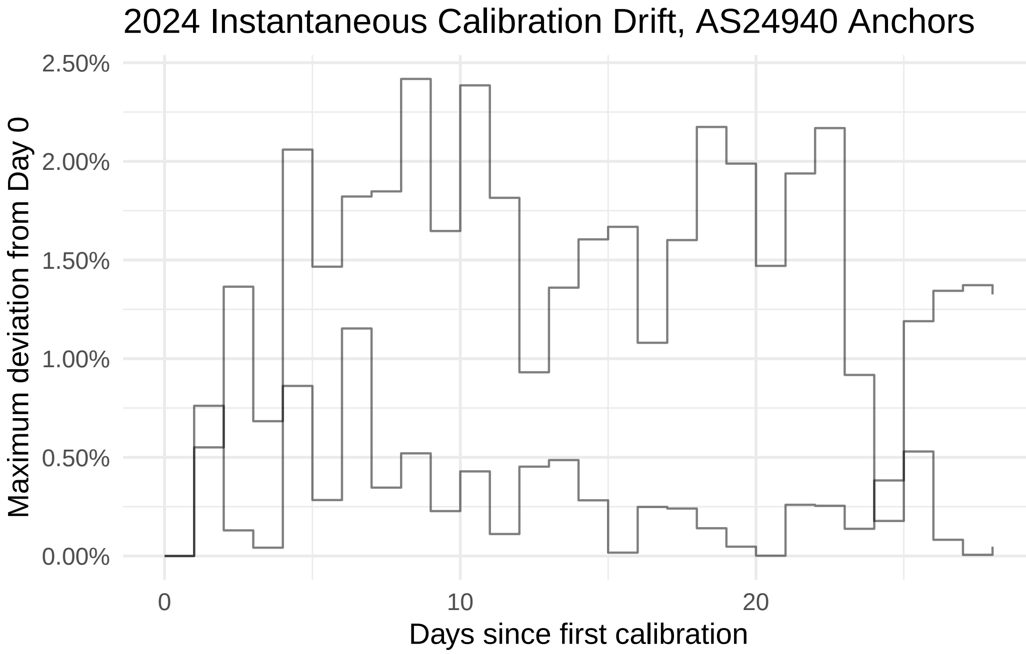
2026 Instantaneous Calibration Drift, AS16347 Anchors
Maximum deviation from Day 0
0.04%
0.02%
0.00%
0
20
40
60
80
Days since first calibration

5.3.5 AS20473

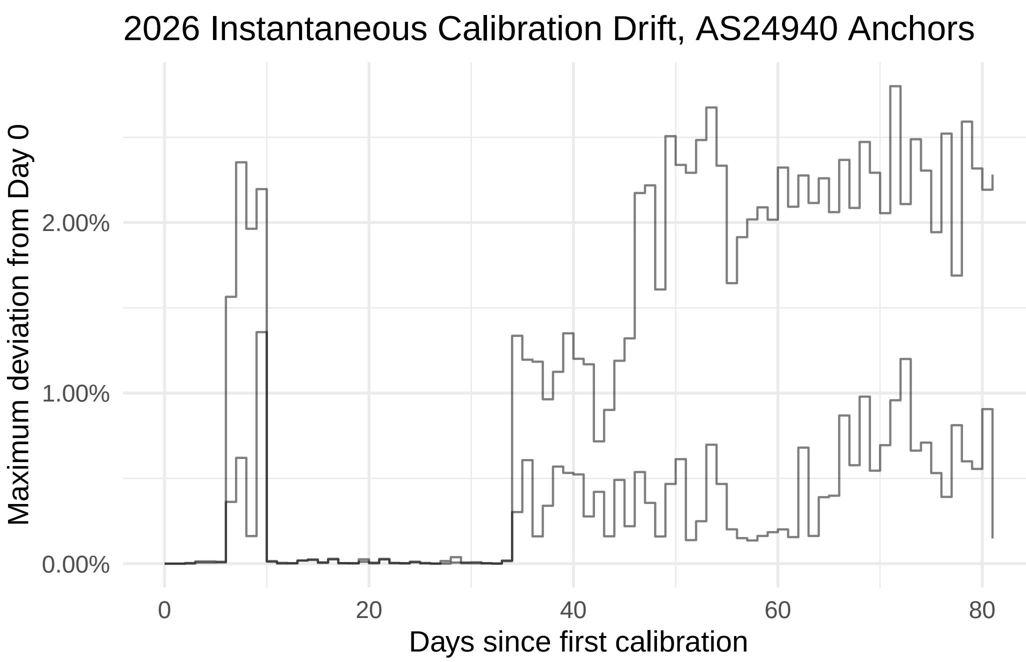

2024 Instantaneous Calibration Drift, AS20473 Anchors
Maximum deviation from Day 0
3.00%
2.00%
1.00%
0.00%
0
10
20
Days since first calibration


2026 Instantaneous Calibration Drift, AS20473 Anchors
Maximum deviation from Day 0
1.00%
0.50%
0.00%
0
20
40
60
80
Days since first calibration

5.3.5 AS24940

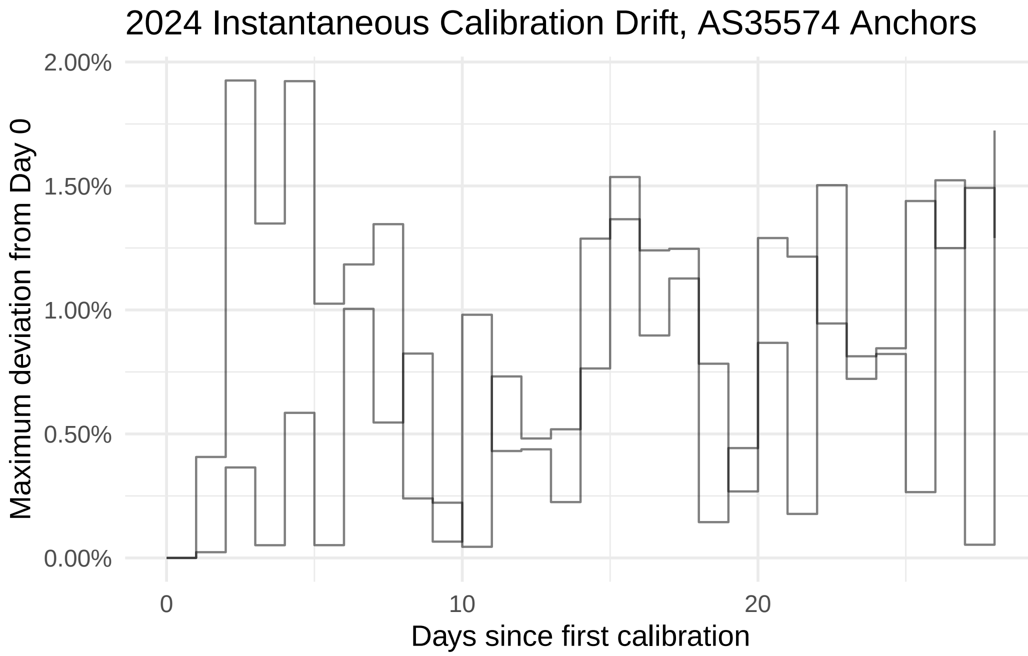


2024 Instantaneous Calibration Drift, AS24940 Anchors
Maximum deviation from Day 0
2.50%
2.00%
1.50%
1.00%
0.50%
0.00%
0
10
20
Days since first calibration

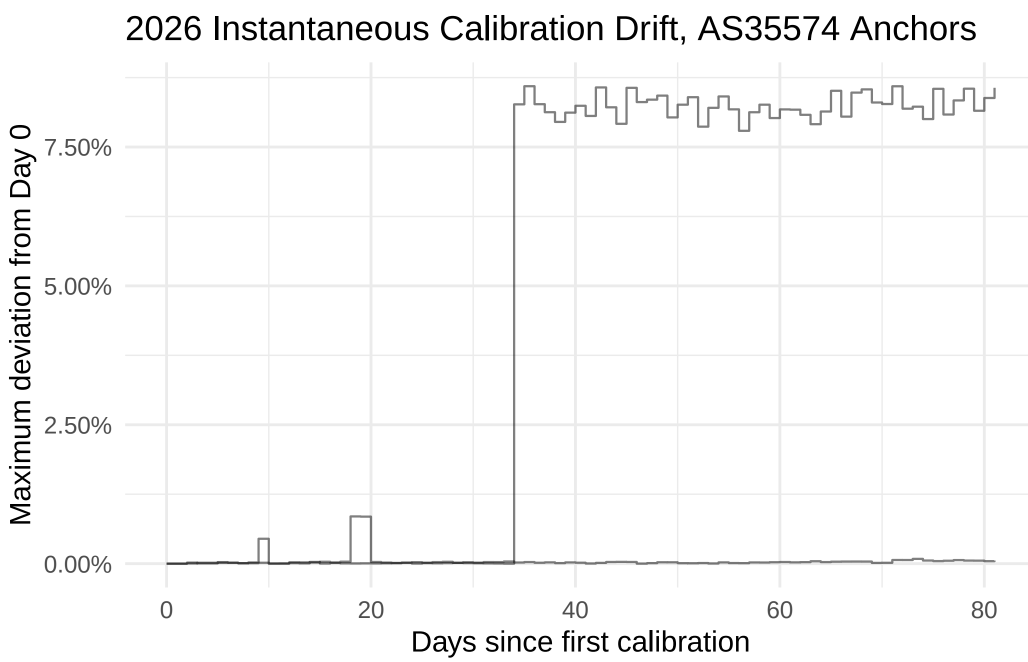


2026 Instantaneous Calibration Drift, AS24940 Anchors
Maximum deviation from Day 0
2.00%
1.00%
0.00%
0
20
40
60
80
Days since first calibration

5.3.5 AS35574

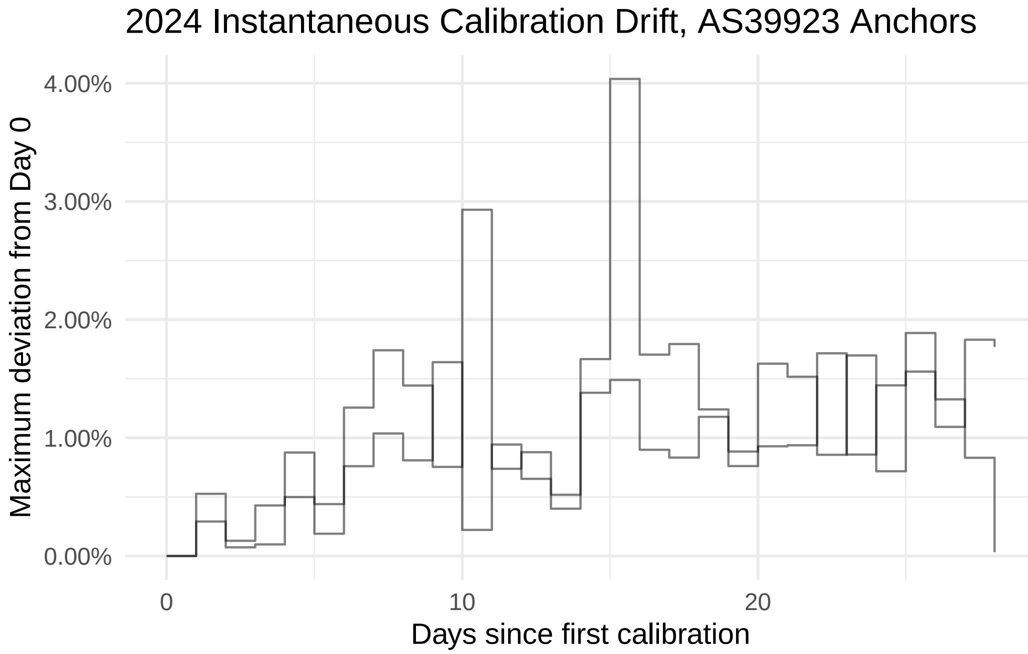


2024 Instantaneous Calibration Drift, AS35574 Anchors
Maximum deviation from Day 0
2.00%
1.50%
1.00%
0.50%
0.00%
0
10
20
Days since first calibration

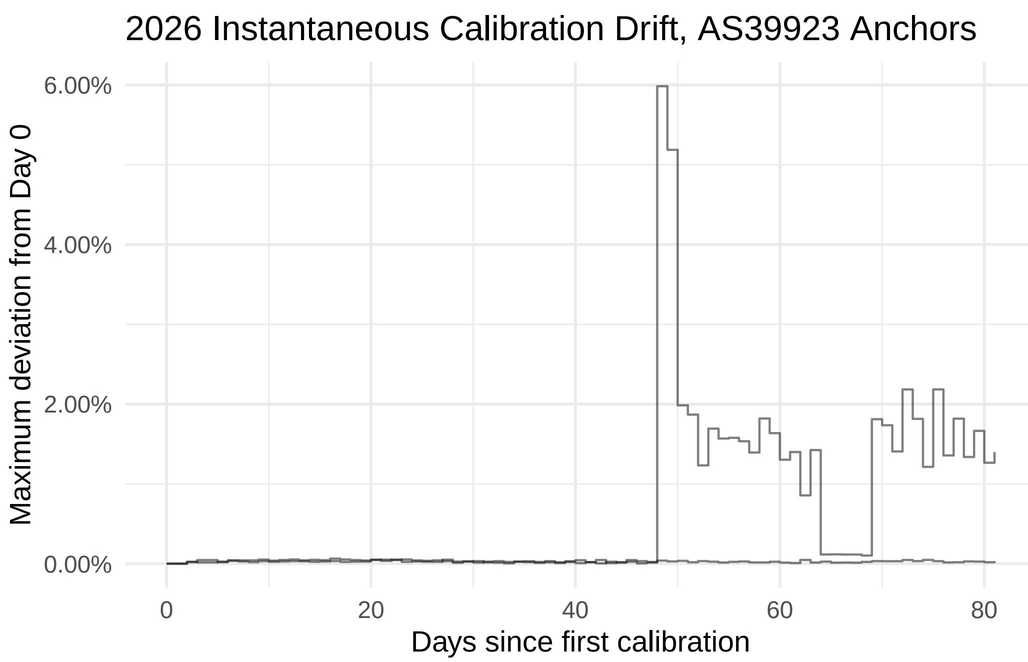


2026 Instantaneous Calibration Drift, AS35574 Anchors
Maximum deviation from Day 0
7.50%
5.00%
2.50%
0.00%
0
20
40
60
80
Days since first calibration

5.3.5 AS39923

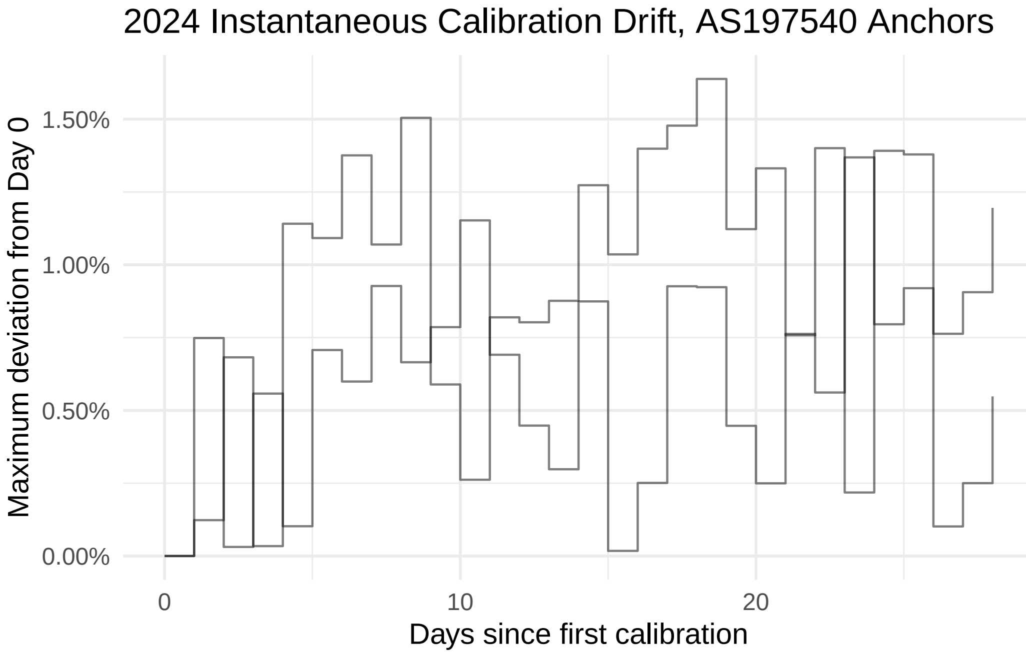


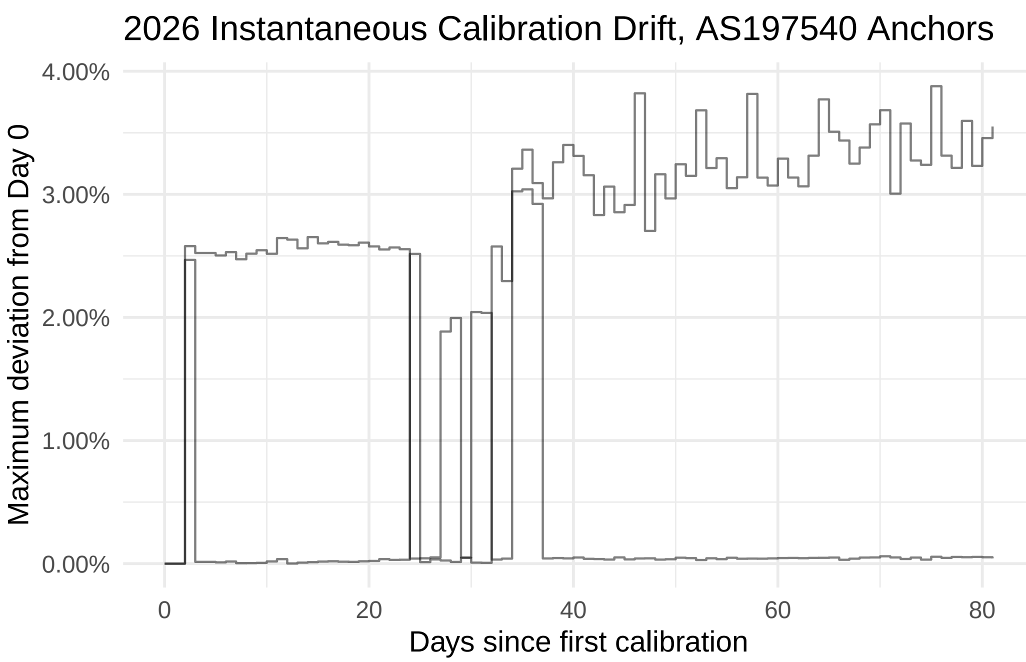
2024 Instantaneous Calibration Drift, AS39923 Anchors
Maximum deviation from Day 0
4.00%
3.00%
2.00%
1.00%
0.00%
0
10
20
Days since first calibration



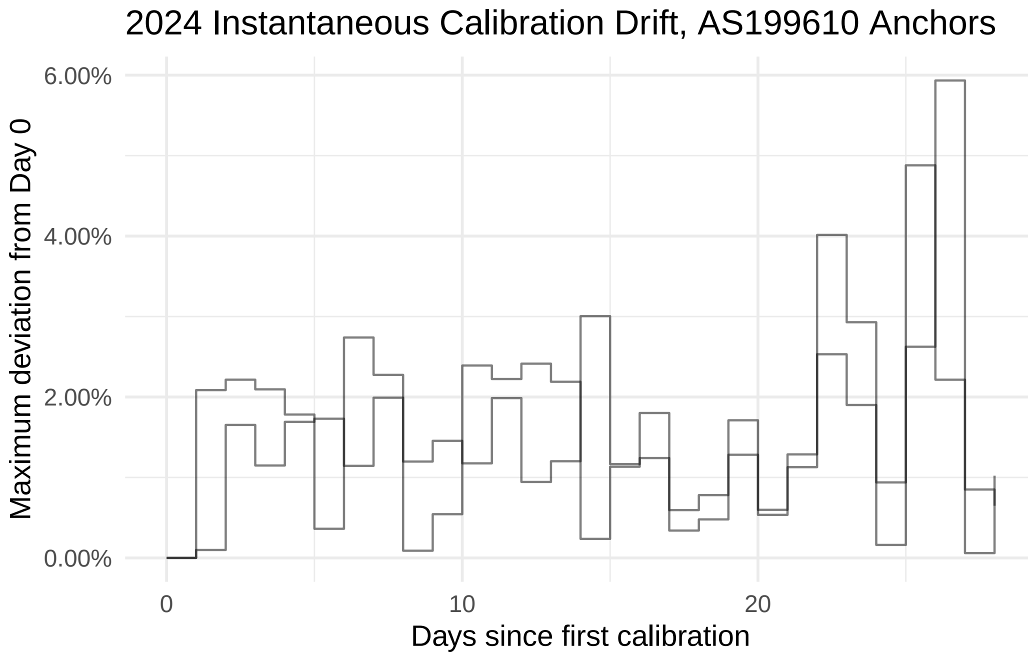
2026 Instantaneous Calibration Drift, AS39923 Anchors
Maximum deviation from Day 0
6.00%
4.00%
2.00%
0.00%
0
20
40
60
80
Days since first calibration

5.3.5 AS8473


2024 Instantaneous Calibration Drift, AS197540 Anchors
Maximum deviation from Day 0
1.50%
1.00%
0.50%
0.00%
0
10
20
Days since first calibration


2026 Instantaneous Calibration Drift, AS197540 Anchors
Maximum deviation from Day 0
4.00%
3.00%
2.00%
1.00%
0.00%
0
20
40
60
80
Days since first calibration

5.3.5 AS199610



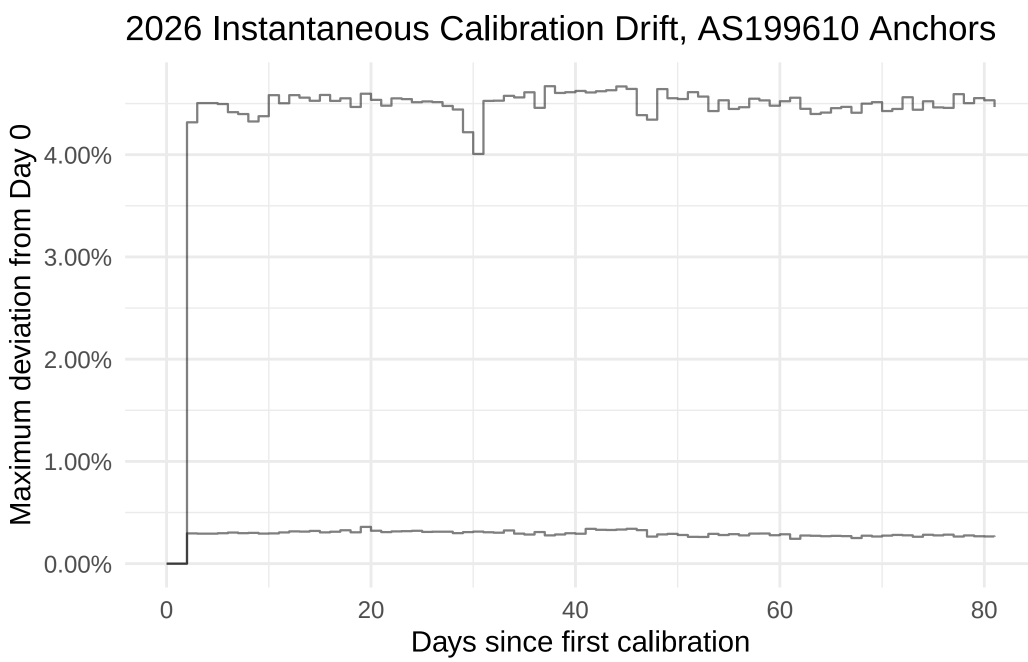


Table 12: AS14061 Anchors

id	in_range_24	total_24	in_range_26	total26	continent	country	city
6968	7	29	4	80	Asia	IN	Bangalore
6967	1	29	49	80	Europe	DE	Frankfurt
6970	1	29	28	80	Asia	SG	Singapore
6971	1	29	4	80	Europe	GB	London

Table 13: AS16276 Anchors

id	in_range_24	total_24	in_range_26	total26	continent	country	city
6800	14	29	1	80	Europe	PL	Ozarów Mazowiecki
6727	5	29	5	80	Oceania	AU	Sydney
6696	1	29	35	80	Europe	FR	Gravelines
6726	1	29	17	80	Asia	SG	Singapore

Table 14: AS3491 Anchors

id	in_range_24	total_24	in_range_26	total26	continent	country	city
7198	24	29	49	80	Asia	AE	Dubai
7078	14	29	70	80	Oceania	AU	Sydney
6779	2	29	1	80	Africa	ZA	Johannesburg
7010	1	29	33	80	Asia	SG	Singapore

Table 15: AS57695 Anchors

id	in_range_24	total_24	in_range_26	total26	continent	country	city
6927	26	29	31	80	North America	US	Secaucus
6928	5	29	1	80	North America	US	Reston
7011	3	29	56	80	North America	US	Seattle

Table 16: AS36236 Anchors

id	in_range_24	total_24	in_range_26	total26	continent	country	city
6379	5	29	1	80	North America	US	Durham

6408	2	29	79	80	North America	US	Phoenix
6487	2	29	1	80	Asia	SG	Singapore

Table 17: AS199524 Anchors

id	in_range_24	total_24	in_range_26	total26	continent	country	city
6668	8	29	1	80	Europe	ES	Madrid
6663	2	29	1	80	Europe	IT	Milano
6644	1	29	18	80	North America	US	Secaucus

Table 18: AS8473 Anchors

id	in_range_24	total_24	in_range_26	total26	continent	country	city
7127	3	29	1	80	Europe	SE	Malmö
7109	2	29	1	80	Europe	SE	Stock-holm

Table 19: AS16347 Anchors

id	in_range_24	total_24	in_range_26	total26	continent	country	city
6443	28	29	1	80	Europe	FR	Maxeville
7048	2	29	1	80	Europe	FR	Nancy

Table 20: AS20473 Anchors

id	in_range_24	total_24	in_range_26	total26	continent	country	city
7037	2	29	1	80	Europe	SE	Stock-holm
6506	1	29	8	80	Asia	SG	Singapore

Table 21: AS24940 Anchors

id	in_range_24	total_24	in_range_26	total26	continent	country	city
6693	3	29	2	80	Europe	DE	Nurem-berg
6332	1	29	5	80	Europe	DE	Nurem-berg

Table 22: AS35574 Anchors

id	in_range_24	total_24	in_range_26	total26	continent	country	city
6450	2	29	1	80	Europe	IT	Rome
6435	1	29	49	80	Europe	IT	Rozzano

Table 23: AS39923 Anchors

id	in_range_24	total_24	in_range_26	total26	continent	country	city
6635	3	29	1	80	Europe	BE	Leuven
6737	1	29	3	80	Europe	BE	Zaven-tem

Table 24: AS197540 Anchors

id	in_range_24	total_24	in_range_26	total26	continent	country	city
6581	1	29	31	80	Europe	DE	Nurem-berg
6799	1	29	29	80	Europe	DE	Nürnberg

Table 25: AS199610 Anchors

id	in_range_24	total_24	in_range_26	total26	continent	country	city
6649	9	29	80	80	North America	US	Los Angeles
6783	3	29	1	80	North America	US	New York City

5.5 Geographic Context of Single Anchor Associated ASNs in the 90-99.9 Range

Due to the generally conservative nature of the IDs associated with ASNs frequently appearing in the 90-999 range, we must not overlook the ASNs making individual appearances. Of course, we maintain AS3941 as being associated with frequent outlier drift behavior amongst the repeated ASNs. We will first take a look at the IDs that are associated with unique ASNs that entered the 90-999 range.

To pare down the IDs for further analysis, we will take away the likely misplaced IDs due to the day 0 ground truth procedure. These IDs will be highlighted in yellow.

Table 26: Anchors Associated with Unique ASNs

asn	id	in_range_24	total_24	in_range_26	total26	country	city
719	6251	1	29	2	80	FI	Tampere
203329	6313	1	29	32	80	DE	Frankfurt am Main
48943	6325	1	29	49	80	AT	Vienna
2108	6327	1	29	30	80	HR	Zagreb
62416	6328	1	29	73	80	PT	Lisbon
1853	6334	1	29	13	80	AT	Vienna
2857	6336	1	29	5	80	DE	Mainz
41000	6423	1	29	2	80	GB	Manchester
7247	6482	1	29	2	80	US	Fremont
31034	6509	1	29	49	80	IT	Arezzo
201126	6516	1	29	2	80	GB	London
39642	6545	1	29	6	80	DK	Kolding
41095	6568	1	29	4	80	SG	Singapore
39617	6674	1	29	3	80	GB	Balderton
24482	6676	1	29	25	80	SG	Singapore
786	6695	1	29	35	80	GB	Didcot
25376	6716	1	29	2	80	GB	Bolton
62310	6751	1	29	5	80	DE	Berlin
41327	6762	1	29	46	80	IT	Palermo
199399	6841	1	29	2	80	GR	Athens
21320	6874	1	29	3	80	FR	Paris
62513	6888	1	29	5	80	CA	Toronto
63949	6918	1	29	6	80	SG	Singapore
39835	6922	1	29	2	80	DE	Göttingen
1213	6965	1	29	2	80	IE	Dublin
39636	7003	1	29	27	80	NL	Arnhem
30971	7008	1	29	2	80	AT	Vienna

13101	7012	1	29	4	80	DE	Kiel
38001	7091	1	29	4	80	SG	Singapore
56381	7097	1	29	53	80	DE	Frankfurt am Main
33920	7125	1	29	32	80	GB	Leeds
29423	7141	1	29	28	80	DE	Frankfurt
50030	7156	1	29	5	80	DE	Berlin
39063	7169	1	29	49	80	DE	Frankfurt
60841	7201	1	29	4	80	NL	Amsterdam
35492	7216	1	29	31	80	AT	Vienna
54316	7224	1	29	9	80	US	Greenwich
29287	7225	1	29	47	80	AT	Vienna
23428	7238	1	29	69	80	US	Kansas City
14907	7261	1	29	2	80	NL	Amsterdam
216444	7272	1	29	13	80	SG	Singapore
32167	7292	1	29	10	80	HK	Hong Kong
34428	7295	1	29	47	80	IT	Carini
50178	6424	2	29	1	80	IT	Meran
60396	6441	2	29	1	80	SE	Lund
39392	6544	2	29	1	80	CZ	Prague
199188	6559	2	29	1	80	GB	London
3320	6724	2	29	1	80	DE	Frankfurt am Main
46997	6794	2	29	78	80	US	Seattle
23483	7052	2	29	1	80	US	Redding
50599	7064	2	29	1	80	PL	Torun
5459	7081	2	29	1	80	GB	London
31510	7083	2	29	1	80	AT	Innsbruck
47295	7096	2	29	1	80	DE	Bautzen
59678	7193	2	29	72	80	US	Piscataway
12110	7199	2	29	25	80	US	Tampa
25248	7211	2	29	2	80	CZ	Prague
13303	7213	2	29	1	80	RS	Nis
41931	7249	2	29	1	80	PL	Bialystok
6724	6331	3	29	1	80	DE	Berlin
39878	6360	3	29	49	80	AT	Salzburg
204092	6494	3	29	1	80	FR	Rennes
59715	6496	3	29	2	80	IT	San Benedetto del Tronto
24971	6809	3	29	3	80	CZ	Brno
29670	6842	3	29	1	80	DE	Berlin
38064	6904	3	29	26	80	NZ	Auckland

6881	7024	3	29	1	80	CZ	Prague
57692	7053	3	29	1	80	FI	Helsinki
267	7105	3	29	1	80	US	Southfield
36444	6389	4	29	1	80	US	Detroit
48362	6476	4	29	1	80	AT	Feldkirch
202297	6547	4	29	1	80	CZ	Prague
28725	6577	4	29	1	80	CZ	Prague
42612	7017	4	29	2	80	ES	Madrid
395823	7043	4	29	79	80	US	Seattle
21034	7110	4	29	5	80	IT	Pescara
40581	7087	5	29	1	80	US	Fayetteville
1280	7253	5	29	1	80	US	Palo Alto
200187	7265	5	29	2	80	DE	Frankfurt am Main
15826	7282	5	29	1	80	FR	Toulouse
29169	6363	6	29	1	80	FR	Saint-Denis
3292	6952	7	29	1	80	DK	Højbjerg
25192	7223	7	29	1	80	CZ	Prague
15954	6604	9	29	1	80	ES	Madrid
24840	7196	9	29	49	80	DE	Frankfurt
39069	6687	10	29	1	80	ES	Murcia
209097	6736	12	29	1	80	FR	Trevoux
15802	6962	12	29	1	80	AE	Dubai
15699	7086	12	29	2	80	ES	Alcalá de Henares
12008	6954	13	29	32	80	IN	Mumbai
202196	6639	14	29	1	80	HK	Hong Kong
201333	6675	14	29	1	80	IT	Piacenza
7752	7221	17	29	77	80	US	Seattle
53698	7159	21	29	78	80	US	Phoenix
206186	6323	25	29	1	80	PL	Katowice

5.6 Percentile Distribution

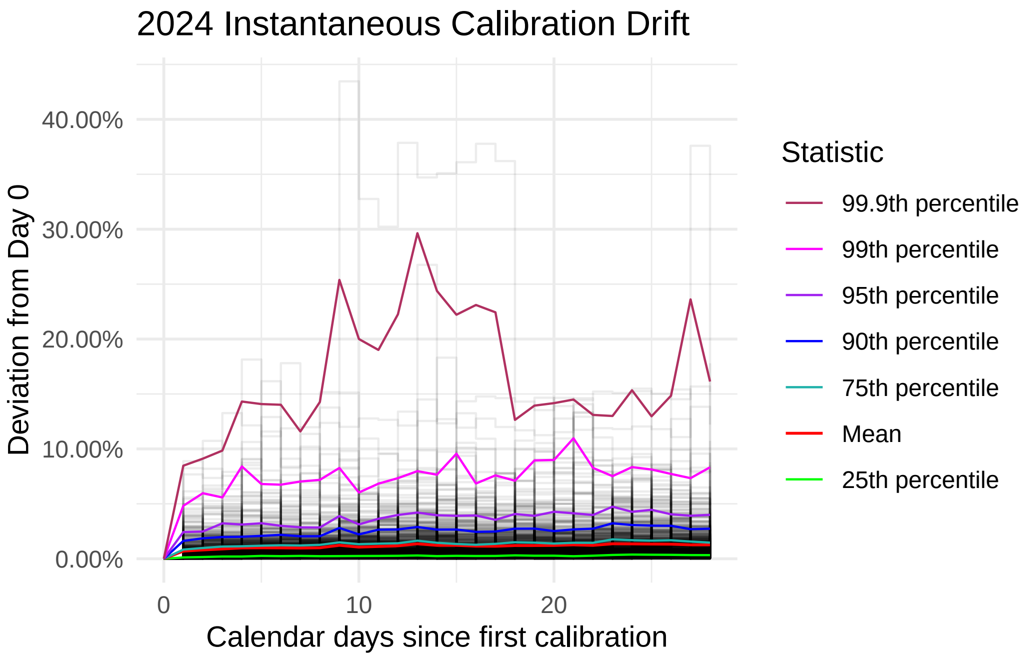
Following Zach's lead, I applied the Instantaneous Calibration Drift visualization on the 2024 and 2026 data. In the 2024 figure, anchor calibration drift is concentrated towards 0% and mainly seems to decrease in density between 2.5% to 7.5% deviation. The remaining fraying seems to happen between 7.5% and 20% deviation primarily, and a few outlier anchors extend above 20%, 30%, and 40%.

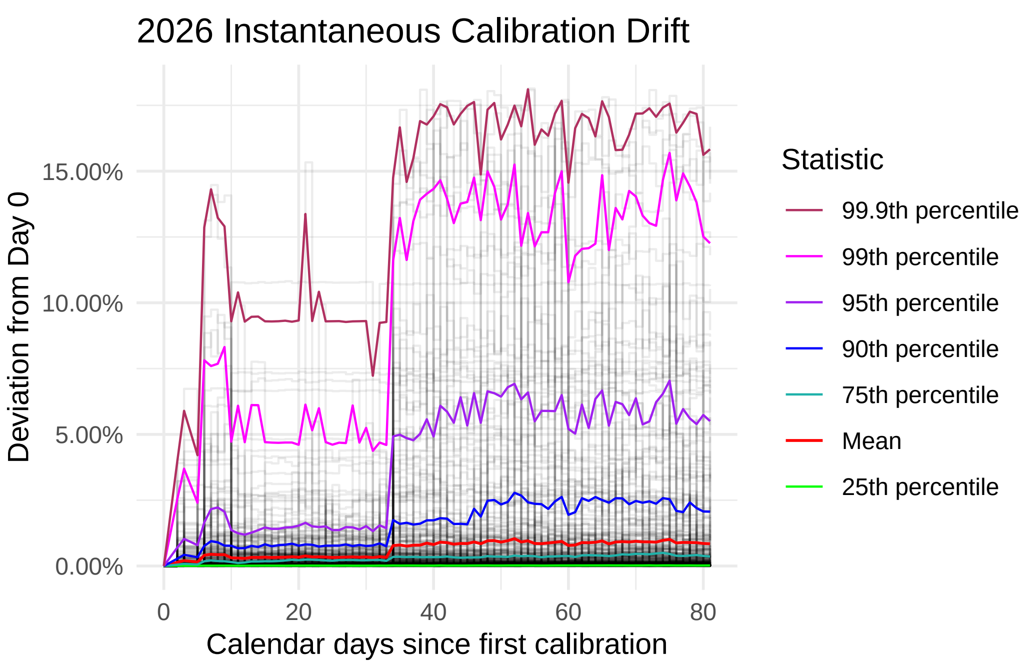
As for the 2026 data, anchor calibration drift seems to be mainly concentrated between 0% and 1%, and spreads similarly to the 2024 2.5% to 7.5% region in the 1% to 2.5% area between day 0 and day 35. This range seems to extend well into 7.5% after day 35. While the outliers drifted between 5% and 15% deviation before day 35, the outlier group is joined by more anchors and extends past 17.5% consistently between day 35 and the end of the recorded period. While the y-axis deviation scale is inconsistent between the two figures, it is clear that the anchor deviation behavior noticeably shifts in the 2026 data.

Due to the significantly lower number of anchors in the 2026 data and the differing time periods, interpreting these figures alone is not enough. If there were 215 or so more anchors in the 2026 data to correspond with the 659 anchors in the 2024 data, perhaps this shift behavior would have been camouflaged by a complement shift down around day 35. It is not clear that the 2024 figure is completely absent of these kinds of shifts either.

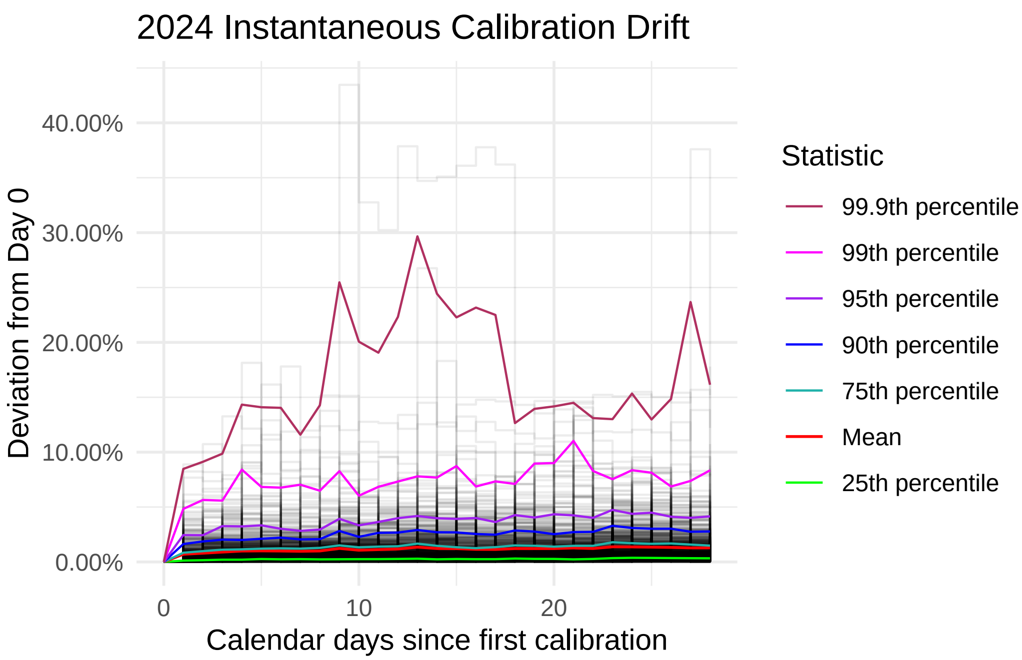
see if the same anchors come up for the 90-999 range

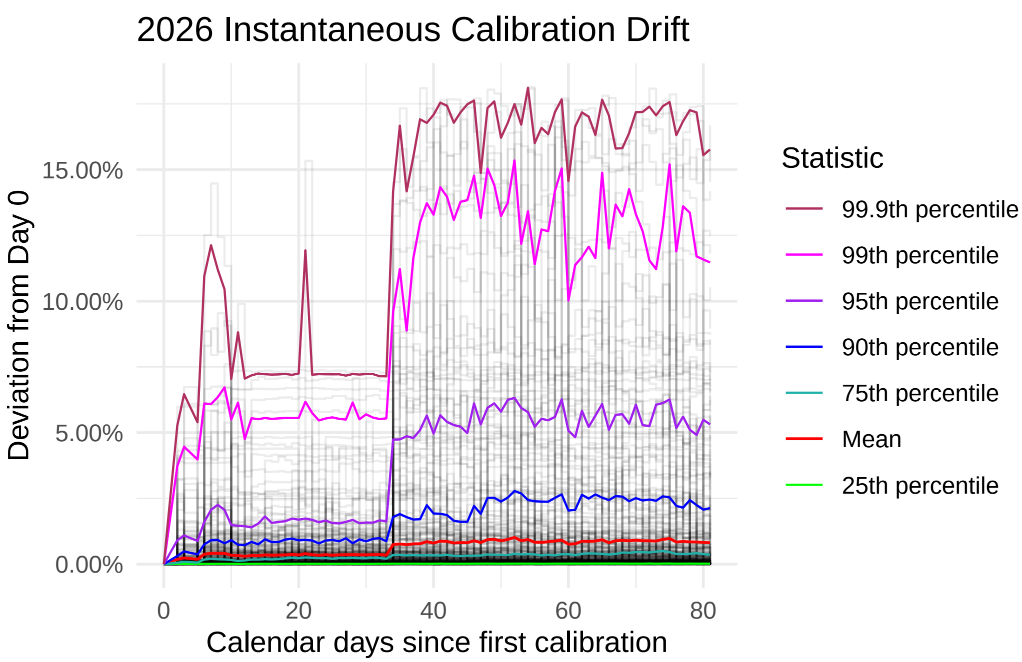
5.7 Instantaneous Drift (after day 0 adjustment + filtering)



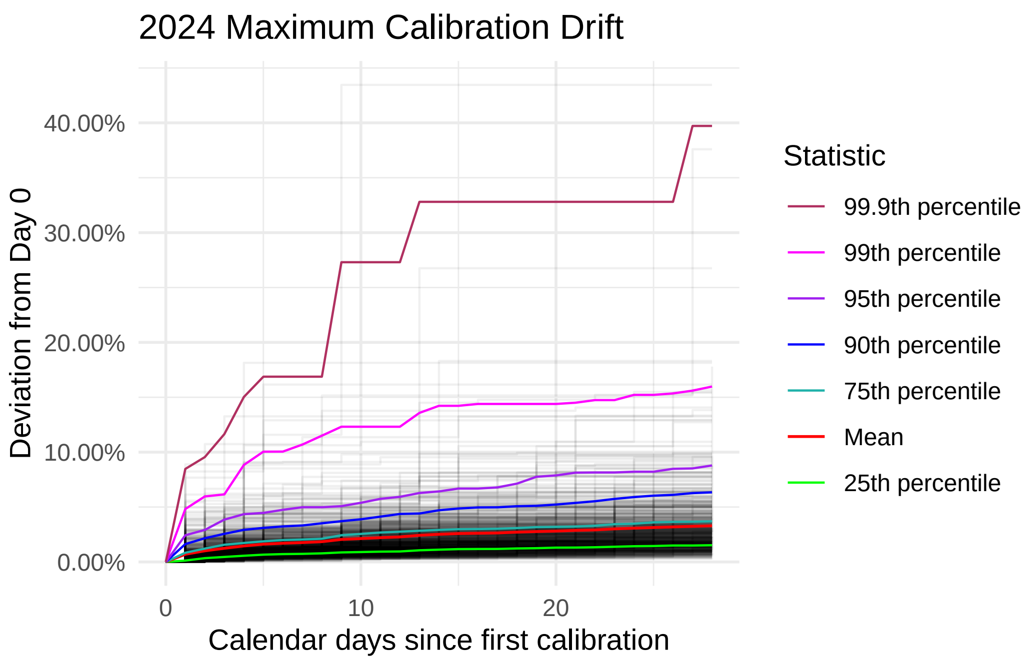


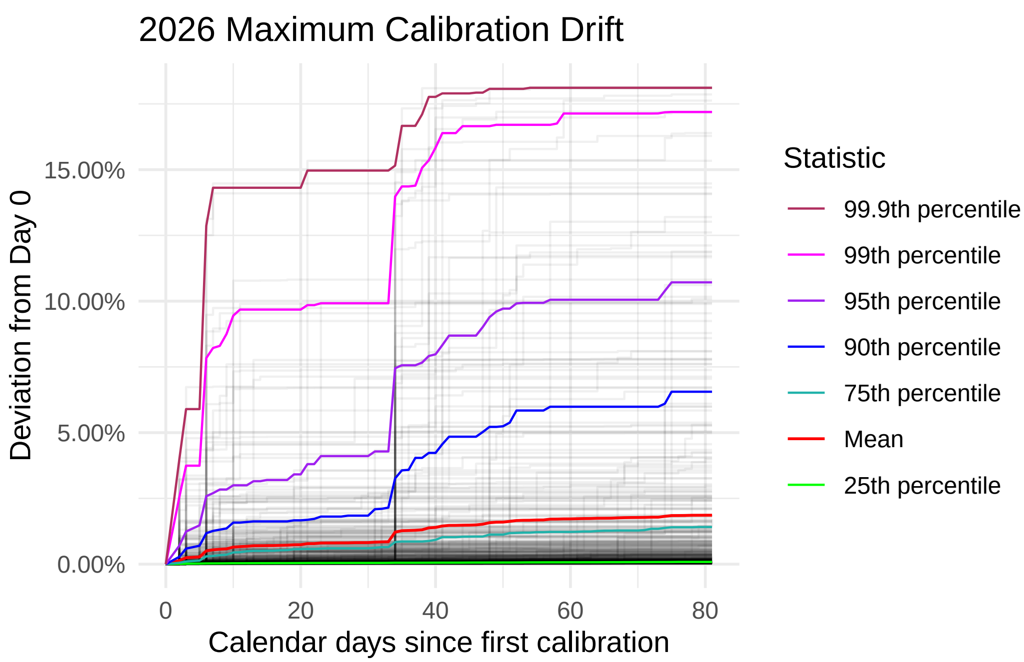
5.8 filter with original part 1 ids filtered



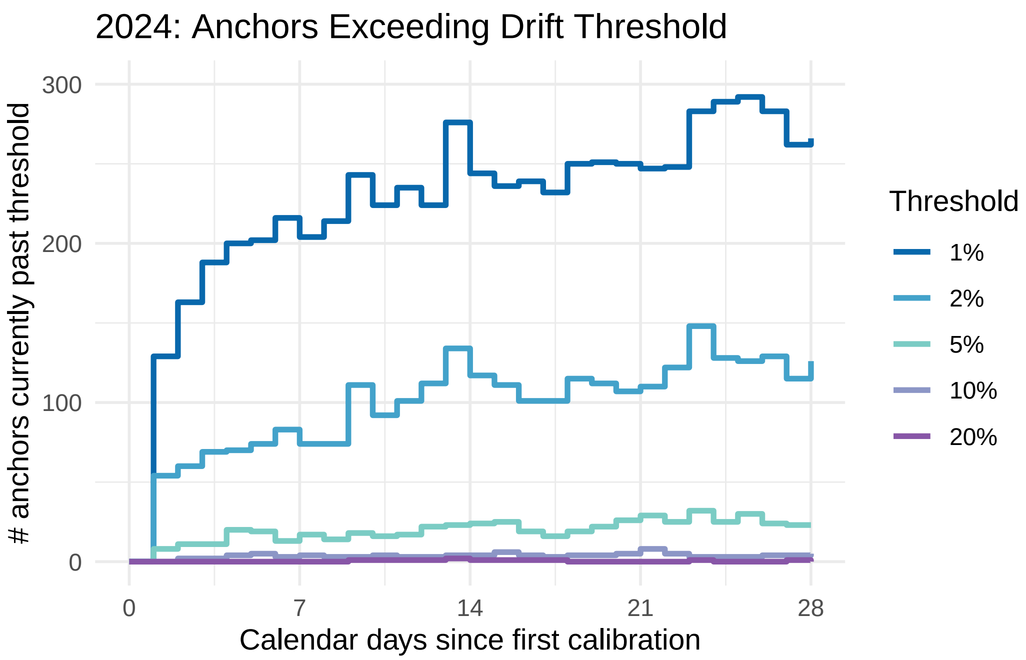


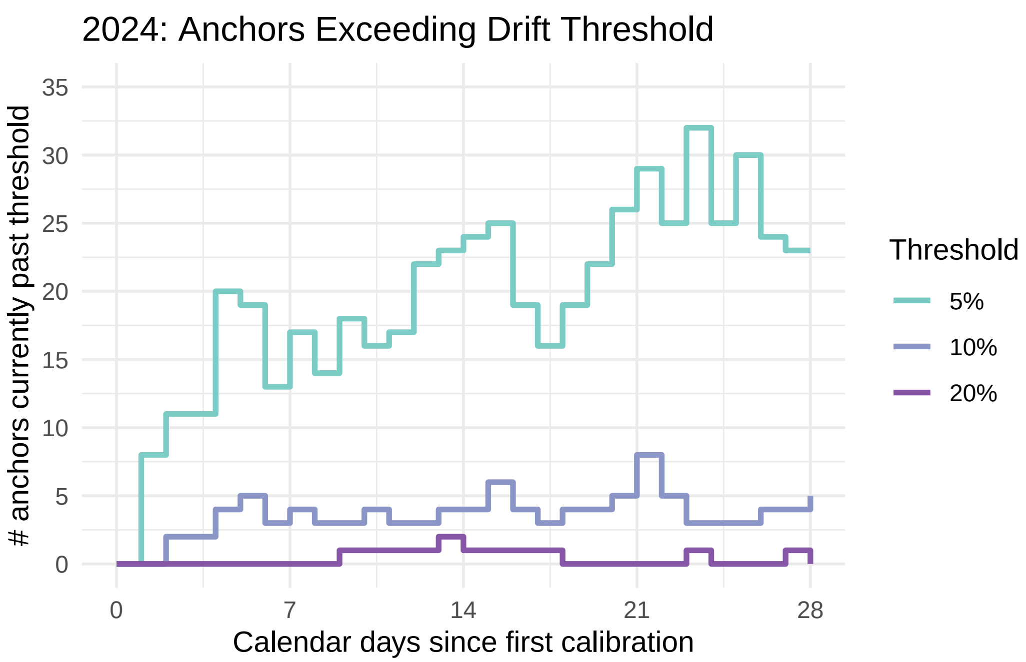
5.9 uhh the first max plot with filtering for the current ids that we chose

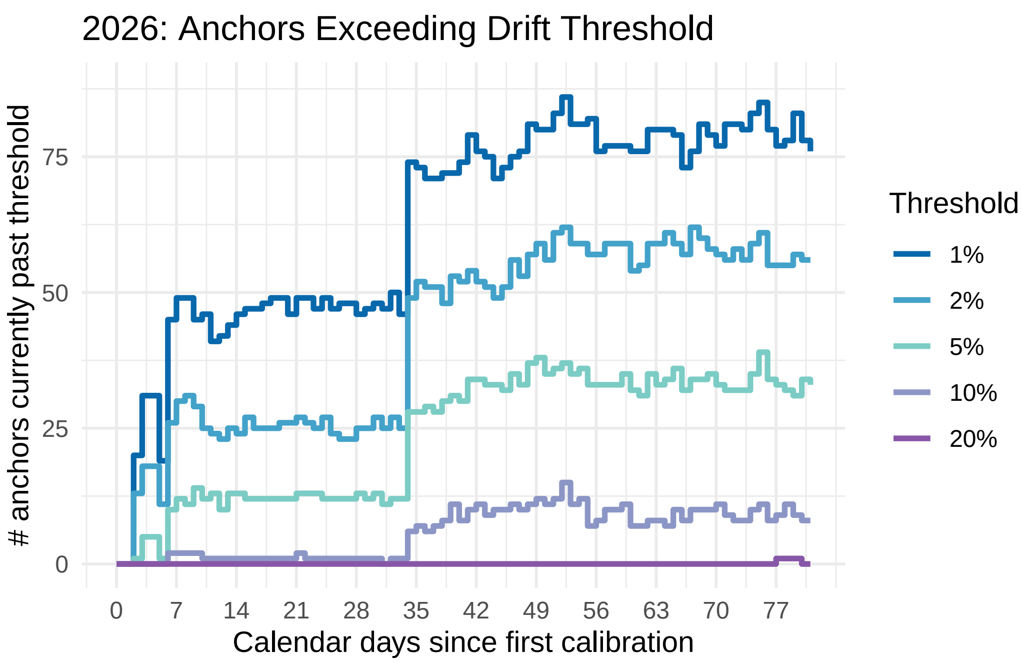


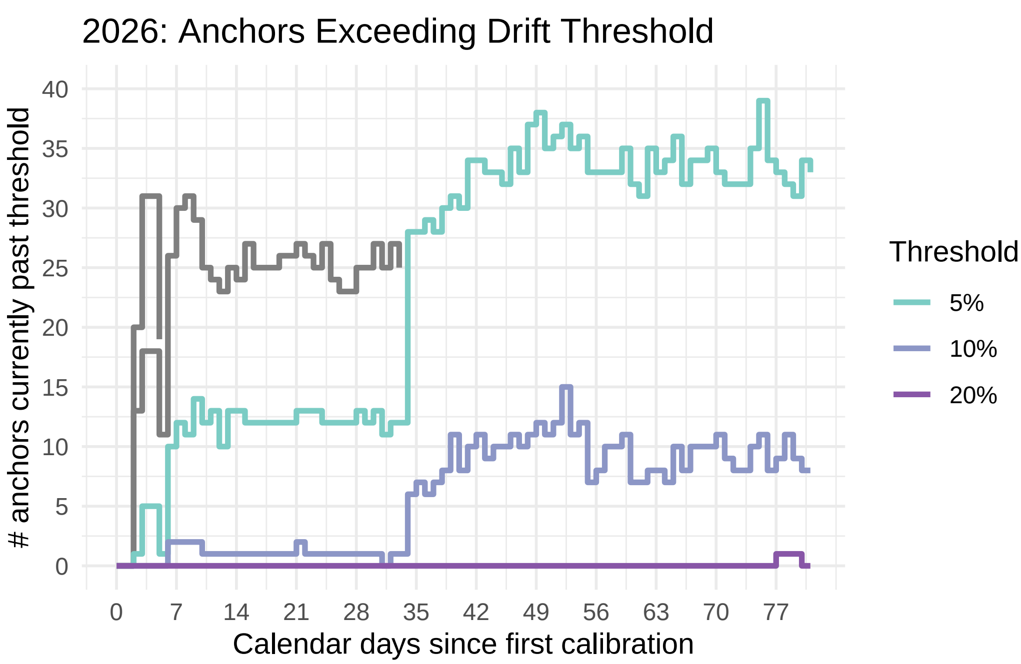


6. Instantaneous Drift Threshold





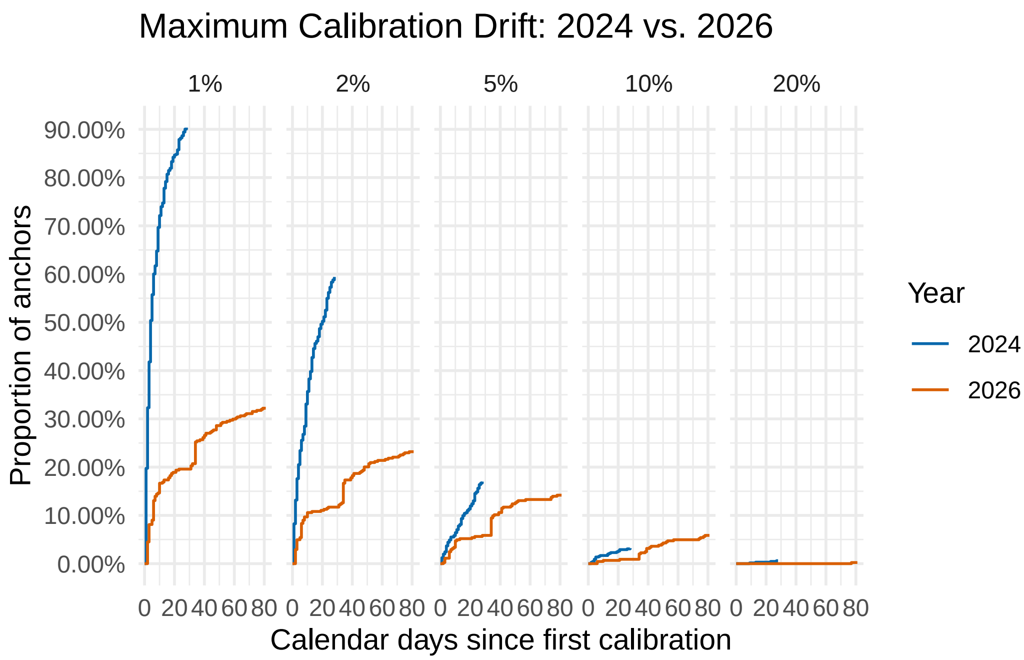

2026: Anchors Exceeding Drift Threshold
anchors currently past threshold
0
25
50
75
0
7
14
21
28
35
42
49
56
63
70
77
Calendar days since first calibration
Threshold
1%
2%
5%
10%
20%


2026: Anchors Exceeding Drift Threshold
anchors currently past threshold
0
5
10
15
20
25
30
35
40
0
7
14
21
28
35
42
49
56
63
70
77
Calendar days since first calibration
Threshold
5%
10%
20%

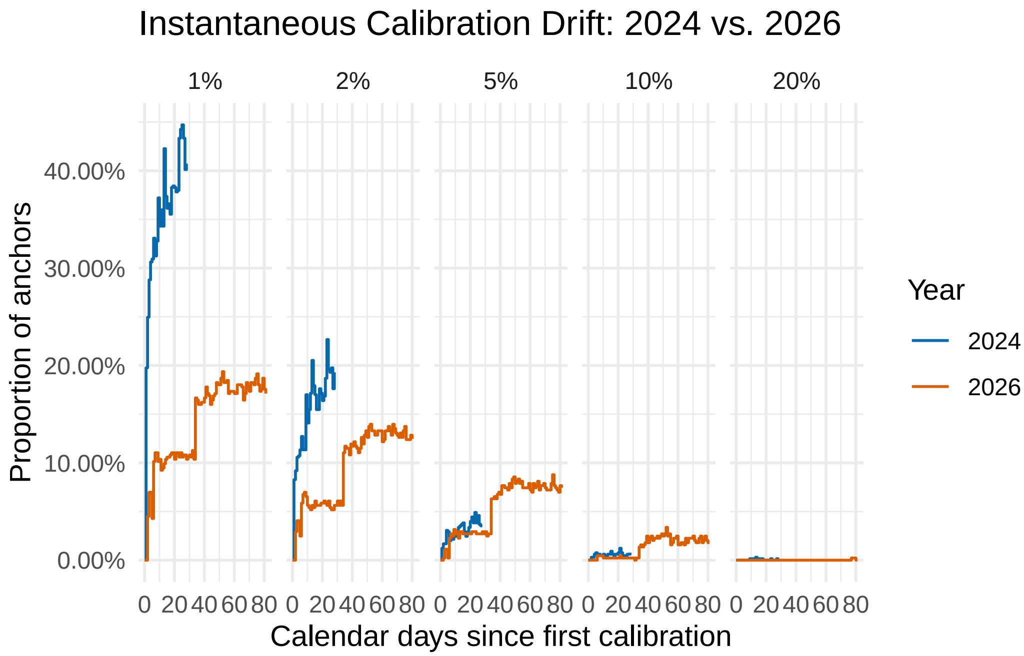
6.1 add mean or median values for each threshold line, note value

7. Direct 2024 vs. 2026 Comparison

For direct comparison, it is useful to put both years onto the same plot. Because the datasets contain different numbers of anchors, percentages are more informative than raw counts.



Look at the curves and see if theres a relationship between the two



Compare proportion of instantaneous drift , calculate relationship between curves

8. Notes for Interpretation

The maximum-drift plots measure whether an anchor has **ever** departed from its initial calibration by a given amount. Consequently, these curves can only increase or remain constant.

The instantaneous-drift plots measure how many anchors are **currently** beyond each threshold. These curves can increase or decrease as anchors move back toward their initial calibration.

A pattern in which maximum drift increases over time while instantaneous drift remains approximately stable suggests that individual anchors experience deviations but that the overall population does not undergo a systematic accumulation of drift.

The 2024 and 2026 comparisons should be interpreted using calendar time. Missing calibration dates in the 2026 dataset remain gaps in the time axis rather than being compressed into consecutive observation numbers.